

Access Control Module

Functional Design & Backend Screens

TICVAI | Professional Reference Document

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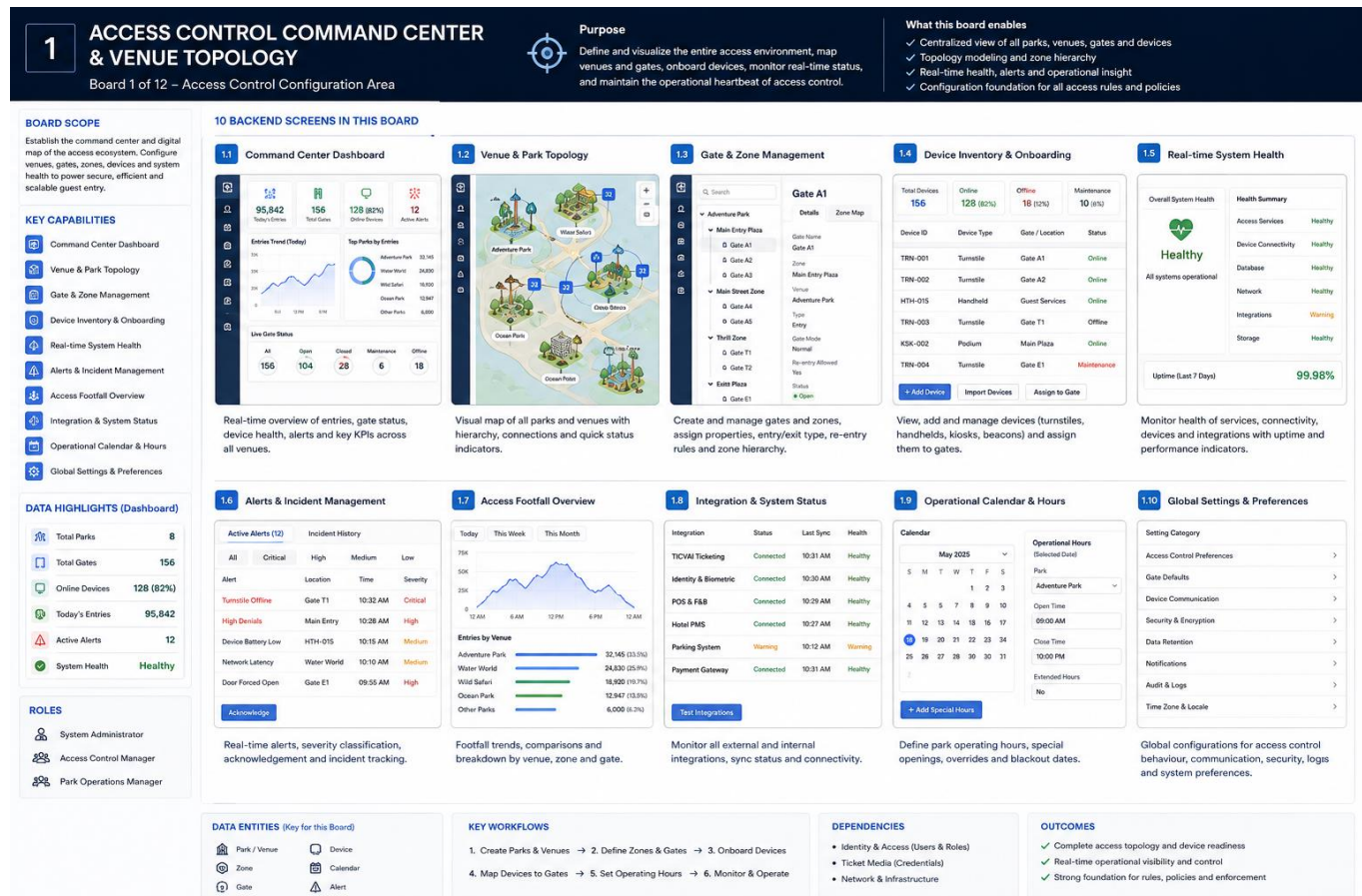
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Board 12 — Final 10-Screen Structure

ACCESS CONTROL — BOARD 1

Access Control Command Center & Venue Topology



Purpose: Establish the physical and logical access-control structure that everything else depends upon.

This board answers:

Where does access happen, how are the physical locations structured, and what access-control infrastructure exists at each location?

It should be highly visual rather than a traditional configuration table.

Screen 1 — Access Control Command Center

Purpose

Central operational/configuration landing page for the complete Access Control module.

Configuration

Display hierarchy such as:

Tenant → Venue → Park → Zone → Attraction → Access Point → Gate → Device

Dashboard should show:

- Total venues
- Active access points
- Entry gates
- Exit gates
- Attraction gates

- Turnstiles
- Handheld devices
- Devices online/offline
- Gates open/closed
- Current in-venue occupancy
- Current admission rate
- Failed scans
- Overrides
- Security alerts
- Offline devices
- Synchronization status

A visual venue map should immediately show gate health using status indicators.

AI Layer

AI Access Operations Assistant highlights:

- abnormal rejection rate
- unusual queue build-up
- offline devices
- abnormal gate traffic
- potential configuration conflicts
- capacity risks
- unusual access patterns

Example:

AI Alert: Gate A03 rejection rate is 18.4%, compared with the normal 2.1%. Most failures are caused by an incorrect access-zone assignment.

Acceptance Condition

Administrator can understand the complete access-control estate and identify operational/configuration problems from one screen.

Screen 2 — Venue & Park Access Structure

Purpose

Define the highest-level physical access hierarchy.

Configuration

Administrator can create:

- Venue
- Park
- Building
- Event space
- Waterpark
- Theme park

- Museum
- Arena
- Stadium
- Exhibition
- temporary venue

Each entity contains:

- Name
- Code
- Tenant
- Parent entity
- Time zone
- Operating calendar
- Capacity
- Access-control enabled
- Default entry policy
- Default exit policy
- Default credential rules
- Offline policy
- Emergency behavior

Support multi-park environments.

Example:

Dubai Entertainment Resort

→ Theme Park

→ Water Park

→ Adventure Park

Acceptance Condition

Any tenant can model single-site or multi-park operations without development.

Screen 3 — Access Area & Zone Builder

Purpose

Divide a venue into controlled access areas.

Configuration

Create:

- Public Zone
- Ticketed Zone
- VIP Zone
- Staff Zone
- Back-of-House
- Attraction Zone

- Restricted Zone
- Fast Pass Zone
- Event Zone
- Temporary Zone

Administrator can drag zones onto a graphical venue structure.

Example:

Main Park

- Public Plaza
- Paid Park Area
- VIP Lounge
- Attraction Cluster A
- Back of House

Each zone receives:

- capacity
- operating schedule
- security classification
- entry requirements
- exit requirements
- allowed credential classes.

Acceptance Condition

Access-controlled zones can be configured and reorganized without software changes.

Screen 4 — Attraction Access Configuration

Purpose

Configure attractions as access-controlled destinations.

Configuration

For each attraction:

- Attraction ID
- Name
- Venue
- Zone
- Capacity
- Entry points
- Exit points
- Fast Pass support
- Height restriction
- Age restriction
- Adult companion requirement
- Membership access
- VIP access

- entitlement requirement
- biometric requirement
- operating calendar
- temporary closure behavior.

Example:**Adventure Coaster****Eligibility:**

Height $\geq$ 130 cm

AND

Valid Park Admission

AND

Ride Entitlement Available

Acceptance Condition

An attraction can independently enforce admission requirements without changing the base ticket product.

Screen 5 — Access Point Directory

Purpose

Create the logical access points where validation occurs.

An **Access Point** is different from a physical reader/device.

Example:**Main Entrance North****could contain:**

Gate N01

Gate N02

Gate N03

Accessible Gate N04

Configuration**Access Point Type:**

- Main Entry
- Exit
- Entry/Exit
- Re-entry
- Crossover
- Attraction Entry
- Fast Pass
- VIP
- Staff
- Accessible/POD
- Group

- Event
- Temporary

Additional settings:

- operating schedule
- allowed direction
- capacity
- default mode
- associated zone
- allowed ticket categories
- fallback behavior.

Acceptance Condition

Operations can logically group multiple physical devices under one controlled access point.

Screen 6 — Gate & Lane Configuration

Purpose

Configure individual physical gates/lanes.

Configuration**Each gate contains:****Identity**

- Gate ID
- Gate name
- Access point
- Location
- lane number

Direction

- Entry
- Exit
- Bidirectional
- Re-entry
- Crossover

Gate type

- Standard
- VIP
- Fast Pass
- Accessible
- Group
- Staff
- Attraction

Operational mode

- Validation
- Free Spin
- Closed
- Emergency / Drop Arm
- Manual
- Count Only.

Allow different gate sizes including larger buggy/wheelchair access gates.

Acceptance Condition

Every physical lane can have independent configuration while inheriting settings from its parent access point.

Screen 7 — Access Control Graphical Map Designer

This should become one of the visually strongest TICVAI screens.

Purpose

Create a graphical digital twin of the access-control environment.

Administrator uploads:

- CAD
- PDF
- image
- venue plan
- architectural drawing.

AI identifies:

- entrances
- exits
- gates
- attractions
- zones
- pathways.

Administrator then drags:

Gate

Turnstile

Scanner

Beacon

Camera

RFID reader

Face recognition device

onto the map.

Clicking a device opens its configuration.

AI

AI can propose:

"I detected 14 likely access lanes and 3 emergency exits. Create the proposed topology?"

Acceptance Condition

Access infrastructure can be configured and monitored geographically from a venue map.

Screen 8 — Access Location Grouping

Purpose

Group multiple access points for operational and capacity purposes.

Example:

Main Entrance

- Gate 01
- Gate 02
- Gate 03
- Gate 04

Adventure Zone

- Coaster Gate
- Drop Tower Gate
- Adventure Hall Gate

The group can calculate:

Entries

Exits

Current Occupancy

Throughput

Capacity

Support nested groups:

Venue → Park → Zone → Attraction → Access Point

Acceptance Condition

Multiple access-control devices can contribute to common occupancy and operational statistics.

Screen 9 — Operating Calendar & Special Access Days

Purpose

Allow access topology and operating behavior to change by date/time.

Configure:

- normal operating days
- weekends
- holidays
- seasonal schedules
- private events
- free-entry days
- maintenance periods

- special events
- ladies-only sessions
- school/group sessions
- after-hours events.

Example:**Free View Day****Main Gate:**

FREE ENTRY

Attraction Gates:

TICKET VALIDATION REQUIRED

The configuration can automatically activate according to calendar rules.

AI**AI warns about conflicts:**

Main Gate is configured for Free Entry on 14 September, but the inherited Park Policy still requires credential validation.

Acceptance Condition

Special-date access configurations can automatically replace standard operation without manual gate-by-gate changes.

Screen 10 — Topology Validation & Publication

Purpose

Final validation and controlled deployment of access configuration.

Before publication, TICVAI automatically validates:

- orphan gates
- devices without access points
- access points without zones
- incorrect entry/exit direction
- missing offline configuration
- conflicting operating calendars
- inaccessible zones
- missing emergency configuration
- capacity inconsistencies
- missing reader/device association
- policy dependencies.

Deployment**Configuration lifecycle:**

Draft → Validate → Test → Approval → Schedule → Publish

Publication targets can be:

- Entire tenant

- Venue
- Park
- Zone
- Access point
- Selected gates/devices.

Support:

Publish Now

or

Schedule Deployment

with rollback to previous configuration.

AI Pre-Publication Check

AI provides a readiness score:

Access Configuration Readiness: 96%

and explains remaining risks.

Acceptance Condition

No access topology enters production without validation, authorization, version tracking and rollback capability.

Board 1 — Final Backend Navigation

The left-side navigation for this board should therefore be:

ACCESS CONTROL

Overview

Access Command Center

Topology

Venue & Parks

Areas & Zones

Attractions

Access Points

Gates & Lanes

Graphical Map

Location Groups

Operations Setup

Operating Calendar

Governance

Validate & Publish

Access Control Module — Board 2

Access Rules & Entitlement Engine

Board 2 of 12 — 10 Backend Configuration Screens

2 ACCESS RULES & ENTITLEMENT ENGINE

Board 2 of 12 — Access Control Configuration

Purpose

Define and configure all access rules, validation logic, entitlements, quantities, journeys, restrictions and consumption behavior that determine whether a guest may enter, exit or access a venue, park, zone or attraction.

What this board enables

- ✓ No-code visual rule engine for all admission policies
- ✓ Configurable entry, exit, re-entry, validity and entitlements
- ✓ Advanced journey, companion, group and multi-park rules
- ✓ Test, simulate, validate and publish rules with confidence

Active Rules: **128**

Draft Rules: **24**

Scheduled: **18**

BOARD OVERVIEW

Configure all access rules, entitlements and validation logic applied at gates and access points.

KEY CAPABILITIES

- Access Rule Command Center
- Visual Rule Builder
- Entry, Exit & Re-entry Rules
- Anti-Passback & Journey Sequence
- Access Validity & Time Rules
- Entitlement Consumption Engine
- Multi-Park & Crossover Rules
- Guest, Companion & Eligibility Rules
- Group Admission & Quantity
- Rule Simulation & Publication

RULE QUICK STATS

- Active Rules: 128
- Draft Rules: 24
- Pending Approval: 12
- Scheduled Changes: 18
- Rules with Conflicts: 6
- Rules Using Biometrics: 42
- Offline Compatible: 116
- Allow Override: 71

AI ASSISTANT

Access Rule Advisor

I can help you build rules, detect conflicts, simulate journeys and recommend improvements.

[Ask AI Assistant](#)

[View AI Insights \(8\)](#)

2.1 Access Rule Command Center

Central repository of all access rules with status, coverage, conflicts and lifecycle management.

Rule Directory

Rule Name	Applies To	Location	Validity	Status	Last Modified
Standard Park Entry	Day Tickets	Adventure Park	Daily	Active	Today 08:12
Annual Pass Entry	Annual Pass	All Parks	Annual	Active	Today 08:45
Fast Pass Silver	Silver Woodcard	Cosplay Drop Tower	3 Uses	Active	Yesterday 18:32
Friday Ladies Night	Event Products	Water Park	Fri PM	Active	Yesterday 10:01
Child Companion Rule	Junior Tickets	Adventure Coaster	Always	Active	2 Days Ago
POD / Nanny Access	POD, Nanny	All Parks	Always	Active	2 Days Ago

[View All Rules](#) [New Rule](#)

2.2 Visual Access Rule Builder

No-code visual builder to create powerful admission rules using a simple WHEN-AT-IF-THEN-AND logic structure.

Rule Name: Adventure Park Entry Rule | Status: Draft

WHEN Credential: Day Ticket | **AT** Location: Adventure Park | **IF** Conditions: Valid Date: Today, Ticket Status: Valid, Guest Category: All | **THEN** Decision: ALLOW ENTRY | **AND** Actions: Consume: 1 Entry, Update: Attendance +1, Record: Scan & Audit

[Preview Logic](#) [Save as Draft](#) [Simulate](#) [New Additional Settings](#)

2.3 Entry, Exit & Re-entry Rules

Configure the complete guest journey lifecycle and quantities for entry, exit and re-entry.

Entry Rules: Max Entries: 1 per day, Entry Reset: Daily, First Entry Required: Yes

Exit Rules: Exit Scan Required: Yes, Max Exits: Unlimited, Exit Before Re-entry: Required

Re-entry Rules: Re-entry Allowed: Yes, Max Re-entries: 1, Re-entry Gate Type: Designated Gate Only, Re-entry Window: Same Day Only

Journey Summary: Entry 1 allowed → Exit Required → Re-entry 1 allowed → Designation Designated Gate

2.4 Anti-Passback & Journey Sequence

Prevent credential sharing and enforce correct entry → exit → re-entry journey sequences.

Anti-Passback Settings: Scope: Per Credential, Minimum Interval: 20 minutes, Applies To: All Gates in Park, Validation Action: Deny Access, Alert Level: High, Notify Security: Yes

Journey Sequence Rules: 1. Entry (Required), 2. Exit (Required), 3. Re-entry (Allowed (1))

[Add Rule to Journey](#)

2.5 Access Validity & Time Rules

Define fixed and dynamic validity including calendars, time windows, grace periods and expiry rules.

Validity Type: Fixed | Relative | Calendar | Time Window

Calendar & Date Rules: Allow on: Selected Days (Mon, Tue, Wed, Thu, Fri, Sat, Sun), Special Dates: 01/01/2025 New Year (Block), 28/05/2025 Maintenance (Block), 15/08/2025 National Holiday (Allow)

Time Window (Optional): Valid From: 15/05/2025 09:00 AM, Valid To: 15/05/2025 11:30 PM, Grace Period: 30 minutes, Expiry after No Show: 20 minutes

[Add Special Date](#)

2.6 Entitlement Consumption Engine

Define entitlements carried by the credential and how they are consumed and reset.

Entitlement List: Add Entitlement: Park Admission, Adventure Coaster, Fast Pass Silver, Meal Voucher, Photo Voucher, Locker

Consumption Configuration: Entitlement Name: Park Admission, Consume Per Access: 1, Reset Type: Daily, Applies To: Adventure Park, Availability: All Entry Gates

Balance Preview: Initial: 1, Consumed Today: 0, Remaining: 1

[Test Consumption](#)

2.7 Multi-Park & Crossover Rules

Configure access across multiple parks with crossover, park limits and prerequisites.

Journey Builder: 1. Adventure Park First Entry, 2. Crossover Same Day, 3. Water Park Second Entry

Journey Summary: 1. Adventure Park Entry (1), 2. Crossover to Water Park (Allowed), 3. Water Park Entry (1), 4. Re-entry after Crossover (No)

[Add Park to Journey](#)

2.8 Guest, Companion & Eligibility Rules

Define guest category rules, height/age restrictions and companion or biometric requirements.

Eligibility Conditions: Guest Category: Child, Require Adult Companion: Yes, Companion Type: Primary Guest, Biometric Validation: Required, Assigned Adult Only: No

Linked Persons (Preview): Primary Guest: Q123456, Child A: Linked, Child B: Linked, POD Companion: Linked, Nanny: Linked

Min Height: 130 cm, **Min Age (Optional)**: 5 Years

2.9 Group Admission & Quantity Validation

Efficient group admission with quantity entry, multiple waves and attendance increment.

Group Ticket Settings: Group Ticket Type: B2B Group Ticket, Authorized Quantity: 50, Multiple Waves: 2

Admission at Gate: Scanned Ticket: 07-2025-000723, Guests Entering Now: 42, Attendance Increment: +12

[Confirm Admission](#)

2.10 Rule Simulation, Conflict Check & Publication

Simulate rules, detect conflicts and publish with versioning, scheduling and rollback capability.

Simulation Panel: Load Scenario: Annual Pass, Credential: Annual Pass, Guest: Adult Member, Park / Venue: Adventure Park, Gate / Access Point: Main Entrance Gate 03

Simulation Result: ALLOW ENTRY, Credential Valid, Valid Date Valid, Park Entitlement OK, Entry Quantity Available, Anti-Passback Passed, Verification Method Valid

Publication: Rule Version: 1.0.0, Publish New Park, Approval Required: Yes, Publish New: Yes

[Publish Rule](#) [Save as Draft](#)

AI INSIGHTS (8)

- 2 Conflicting Rules: Need attention
- 3 Unused Rules: Consider archiving
- 1 Overly Broad Rule: Review override settings
- 2 Missing Gate Coverage: Check zone mapping

[View All Insights](#)

KEY WORKFLOWS

Create Rule → Build Logic → Set Conditions → Define Actions → Simulate → Approve → Publish

DEPENDENCIES

- Board 1 - Venue Topology, Gates, Zones
- Products & Entitlements (Pricing)
- Ticket Media (Credentials)
- Device Configuration (Board 6)

OUTCOMES

- ✓ Consistent and secure admission across all gates
- ✓ Configurable without development
- ✓ Reduced fraud and credential sharing
- ✓ Better guest experience and operational efficiency

Board 1 established **where access happens**: venue → park → zone → attraction → access point → gate.

Board 2 now defines **whether access should be granted and what happens when the credential is scanned**.

This is one of the most important configuration boards in the entire Access Control module because the matrix requires configurable rules for areas/attractions, number of entries/exits/re-entries, anti-passback, required entry/exit sequences, fixed and dynamic validity, adult companion requirements, biometrics, overrides, groups, multi-park crossover, and entitlement consumption—all configurable without development.

Core design principle

The engine should not create separate hard-coded logic for every ticket type.

Instead:

Credential + Entitlements + Access Context + Access Policy → Decision + Action

For example:

Annual Pass + Adventure Park Gate + Saturday 14:30 + Valid Membership + Entry Available → **ALLOW ENTRY**

or:

Day Ticket + Re-entry Gate + No Previous Exit → **DENY — EXIT REQUIRED BEFORE RE-ENTRY**

Screen 2.1 — Access Rule Command Center

Purpose

Central configuration dashboard for all access and entitlement rules.

Dashboard KPIs

Show:

- Active Access Rules
- Draft Rules
- Scheduled Rules
- Rules Pending Approval
- Venues Covered
- Products/Tickets Covered
- Rules with Conflicts
- Rules Using Biometrics
- Rules Allowing Override
- Offline-Compatible Rules
- Recently Modified Rules
- Upcoming Rule Changes

Rule Directory

Each rule displays:

Rule	Applies To	Location	Validity	Status
Standard Park Entry	Day Tickets	Adventure Park	Daily	Active
Annual Pass Entry	Annual Pass	All Parks	Annual	Active
Fast Pass Silver	Silver Wristband	Selected Rides	3 uses	Active
Friday Ladies Night	Event Products	Water Park	Friday PM	Active
Child Companion	Junior Tickets	Selected Attractions	Always	Active

Filters:

Venue / Park / Attraction / Product / Credential / Rule Type / Status / Date

AI Assistant

AI identifies:

- duplicate rules
- conflicting rules
- unused rules
- risky overrides
- unusual restrictions
- missing access coverage.

Acceptance Condition

Administrator can centrally find, understand, create, clone, modify and govern all admission rules.

Screen 2.2 — Visual Access Rule Builder

Purpose

Provide a **no-code rule engine**.

This is where TICVAI should become significantly easier to configure than traditional access-control systems.

Visual Structure

WHEN

Guest presents:

Ticket / Membership / Pass / Credential

AT

Venue → Park → Zone → Attraction → Gate

IF

conditions are satisfied

THEN

Allow / Deny / Refer to Operator / Override Eligible

AND

perform actions.

Example

WHEN

Product = Adventure Day Pass

AT

Adventure Park / Main Entrance

IF

Visit Date = Today

AND Ticket Status = Valid

AND Remaining Entry > 0

THEN

ALLOW ENTRY

AND

Consume 1 Entry

Update Attendance +1

Record Scan

Support nested:

AND / OR / NOT

logic.

AI**Administrator could type:**

"Allow annual-pass holders into Adventure Park twice per day and permit one re-entry if they previously exited."

AI builds the proposed rule visually for administrator review.

Acceptance Condition

Complex admission rules can be configured without code or vendor development.

Screen 2.3 — Entry, Exit & Re-entry Rules**Purpose**

Configure admission quantity and journey sequencing.

The source matrix explicitly requires configurable quantities for entry, exit and same-day re-entry, anti-passback intervals, required exit before re-entry, and required entry before exit.

Configuration**Entry**

- Unlimited
- Once
- N times
- N per day
- N per period

Exit

- Exit scan required
- Exit scan optional
- Unlimited exit
- N exits

Re-entry

- Allowed / Not Allowed
- Maximum re-entries
- Same-day only
- Designated gate required
- Exit required first
- Re-entry window

Example:**Standard Day Ticket**

Entry: 1

Exit: Required

Re-entry: 1

Re-entry Gate: Designated Gate Only

Acceptance Condition

The complete entry → exit → re-entry lifecycle can be configured independently by ticket/product/access policy.

Screen 2.4 — Anti-Passback & Journey Sequence

Purpose

Prevent credential sharing and impossible access sequences.

Anti-Passback Configuration

Administrator sets:

Minimum interval between scans

Example:

20 minutes

Scope:

- Credential
- Guest
- Gate
- Attraction
- Park
- Venue

Journey Sequence

Configure required sequences such as:

ENTRY → EXIT → RE-ENTRY

or:

PARK A → CROSSOVER → PARK B

or:

MAIN ENTRY → ATTRACTION ENTRY

Violation Action

If sequence is violated:

- Deny
- Warning
- Refer to operator
- Require supervisor
- Allow override
- Trigger security alert.

Example

Guest scans the same annual pass at Gate A and then someone scans it 45 seconds later at Gate B.

Result:

RED — ANTI-PASSBACK VIOLATION

Acceptance Condition

The system prevents credential sharing and invalid journey sequences while allowing configurable operational exceptions.

Screen 2.5 — Access Validity & Time Rules

Purpose

Determine **when** access is permitted.

Fixed Validity

Support:

- From / To
- N days after sale
- N days after activation
- N days after first use
- End of Day
- End of Week
- End of Month
- End of Year.

Calendar Validity

Configure:

- weekdays
- weekends
- peak dates
- off-peak dates
- holidays
- seasons
- event dates
- blackout dates.

Time Windows

Example:

Morning Ticket

09:00–13:00

Grace Period:

30 minutes

No-Show / Expiry

The matrix also requires a configurable expiry when a credential is not used within a specified period relative to its admission/performance time.

Example:

Performance: 14:00

Admission tolerance: 20 min

Automatically expire at:

14:20

Acceptance Condition

Access validity can be controlled by dates, relative periods, calendars and precise time windows.

Screen 2.6 — Entitlement Consumption Engine**Purpose**

Determine **what gets consumed when access is granted**.

This is critical because one credential may represent several different entitlements.

The matrix specifically describes a single QR capable of carrying park admission, ride entitlement, meal voucher, coupon, photo voucher and re-entry rights.

Entitlement Types**Support:**

- Park Admission
- Attraction Admission
- Ride
- Fast Pass
- Meal
- Voucher
- Photo
- Locker
- Event
- Experience
- Re-entry
- Membership benefit
- custom entitlement.

Consumption Rules**For each entitlement:****Quantity**

3

Consumption per validation

1

Reset

Daily

Location

Selected Attractions

Example — Silver Fast Pass**Initial:****3 Fast Passes**

Ride 1 → Remaining **2**

Ride 2 → Remaining **1**

Ride 3 → Remaining **0**

The source also requires configurable unlimited Gold Fast Pass behavior and the ability to restrict usage to one access per ride.

Acceptance Condition

A validation event can consume the correct entitlement and immediately calculate the remaining balance.

Screen 2.7 — Multi-Park & Crossover Rules

Purpose

Configure complex access between multiple parks/venues.

The matrix specifically requires multi-park access on different days, same-day crossover, park-specific entry quantities and conditional access based on previous park admission.

Visual Journey Builder

Example:

2-Park Hopper

DAY 1

Adventure Park

↓

First Entry

↓

Crossover Allowed

↓

Water Park

Configure:

- allowed parks
- park order
- same-day crossover
- different-day access
- number of park entries
- crossover quantity
- crossover time
- prerequisite park
- re-entry after crossover.

Important

TICVAI should distinguish:

ENTRY

RE-ENTRY

CROSSOVER

because they have different operational meanings and should be reported separately.

Acceptance Condition

Complex multi-park journeys can be configured without creating separate ticket products for every possible journey.

Screen 2.8 — Guest, Companion & Eligibility Rules

Purpose

Apply access conditions based on guest characteristics and relationships.

Eligibility

Configure:

- Adult
- Child
- Junior
- Senior
- POD
- POD Companion
- Nanny
- VIP
- Member
- Staff
- Accreditation
- Customer Segment.

Child Protection

Support:

Child Ticket + Assigned Adult Ticket

or:

Child + Biometric Verification of Assigned Adult
before admission/exit.

The source explicitly requires both adult-ticket pairing and biometric validation of the assigned adult for child journeys.

Companion Relationship

Example:

Primary Guest

Q123456

Linked:

- Child A
- Child B
- POD Companion

- Nanny

Conditions

Example:

IF

Guest Category = Child

AND

Attraction = Adventure Coaster

THEN

Assigned Adult Required

Acceptance Condition

Access rules can consider guest category, linked persons and companion requirements.

Screen 2.9 — Group Admission & Quantity Validation

Purpose

Handle B2B groups, school groups, tour groups and family/group tickets efficiently.

The matrix specifically calls for faster admission for large B2B groups and the ability for one QR/group ticket to represent multiple admissions.

Configuration

Group ticket:

Authorized Quantity: 50

At gate, scanner displays:

GROUP TICKET

Authorized Guests: **50**

Operator enters:

Guests Entering Now: 42

System records:

Entered: **42**

Remaining: **8**

Attendance increment:

+42

Admission Modes

Support:

- Entire Group
- Partial Group
- Multiple Waves
- Leader + Guests
- Single QR Multi-Entry

- Individual Child Tickets under Group
- Prevalidated B2B Manifest.

Fast Group Mode

For large B2B arrivals:

Scan Group Leader QR → Select/Confirm Attendance → Open Group Lane

This directly addresses the source requirement to improve entrance flow where multiple tickets may be stored on one device.

Acceptance Condition

Large groups can be admitted quickly without scanning every guest individually when the configured product permits group admission.

Screen 2.10 — Rule Simulation, Conflict Check & Publication

Purpose

No access rule should reach a live gate without being tested.

Simulation

Administrator selects:

Credential

Annual Pass

Guest

Adult Member

Date

Friday

Time

18:30

Location

Water Park

Gate

Main Gate 03

Previous Journey

Adventure Park entered at 14:00

Exited at 17:30

Then click:

RUN SIMULATION

TICVAI evaluates every rule.

Decision Trace

- ✓ Credential Valid
- ✓ Visit Date Valid
- ✓ Water Park Entitlement
- ✓ Crossover Allowed

- ✓ Previous Park Requirement Met
- ✓ Entry Quantity Available
- ✓ Anti-Passback Passed
- ✓ Verification Method Valid

FINAL DECISION:

ALLOW

For denial:

DENY — RULE AC-284

and show exactly which condition failed.

Conflict Detection

AI detects:

- contradictory rules
- overlapping rules
- unreachable rules
- duplicate conditions
- missing gate coverage
- impossible journey logic
- overly broad overrides
- offline-incompatible conditions.

Governance

Lifecycle:

Draft → Simulate → Validate → Approval → Schedule → Publish

Support versioning:

Rule V1.0

Rule V1.1

Rule V2.0

and:

Rollback

Acceptance Condition

Every access rule can be simulated, validated, approved, versioned, published and rolled back with a complete audit trail.

Board 2 — Final Screen Structure

#	Backend Screen	Core Responsibility
2.1	Access Rule Command Center	Central access-rule management

#	Backend Screen	Core Responsibility
2.2	Visual Access Rule Builder	No-code admission logic
2.3	Entry, Exit & Re-entry Rules	Admission quantities and lifecycle
2.4	Anti-Passback & Journey Sequence	Sharing prevention and sequence control
2.5	Access Validity & Time Rules	Dates, calendars, time and expiry
2.6	Entitlement Consumption Engine	Consume and track access benefits
2.7	Multi-Park & Crossover Rules	Cross-park admission journeys
2.8	Guest, Companion & Eligibility Rules	Child, companion and guest eligibility
2.9	Group Admission & Quantity Validation	B2B/group admission
2.10	Rule Simulation, Conflict Check & Publication	Testing, governance and deployment

Board 2 key workflow

Select Credential/Product → Select Access Location → Build Conditions → Configure Entry/Exit/Re-entry → Configure Validity → Define Entitlement Consumption → Add Guest/Group Conditions → Simulate → Approve → Publish

This board becomes the **core admission decision engine of TICVAI**. Board 3 can then focus specifically on **Dynamic QR & Secure Digital Credentials**, including rotating QR, device binding, BLE/geofencing, screenshot protection, offline cryptographic validation and secure credential activation.

Access Control Module — Board 3

Dynamic QR & Secure Digital Credentials



Board 3 of 12 — 10 Backend Configuration Screens

Board 3 is the **security and activation layer** for digital access credentials.

Board 1 defined **where access happens**. Board 2 defined **whether the guest is entitled to access**. Board 3 defines **how the digital credential itself is securely generated, activated, refreshed, bound, protected, validated offline, and invalidated**.

This board directly covers the Dynamic QR requirements in **3.1.1–3.1.10**, including registration, blurred/inactive QR states, proximity activation, dynamic refresh, device binding, transfer, offline cryptographic validation, BLE/geofencing, suspicious-use detection, scan history, and embedded offline entitlements.

Screen 3.1 — Digital Credential Security Command Center

Purpose

Central configuration and monitoring page for all secure digital credentials.

Credential Portfolio

Show:

- Dynamic QR Tickets
- Membership Credentials
- Annual Passes

- Mobile Wallet Credentials
- Loyalty Credentials
- Digital Passes
- Event Credentials

KPI Cards

- Active Digital Credentials
- Dynamic QR Enabled
- Device-Bound Credentials
- Location-Protected Credentials
- Offline-Ready Credentials
- Credentials Revoked Today
- Transfer Events
- Security Alerts
- Suspicious Sessions

Security Health

For each credential profile:

Credential	Dynamic	Device Bound	Location	Offline	Security
Day Ticket	✓	✓	Gate	✓	Strong
Annual Pass	✓	✓	Venue	✓	Strong
Member Pass	✓	✓	Venue	✓	Strong
Event Ticket	✓	Optional	Event	✓	Standard

AI Security Assistant

AI identifies:

- weak credential configurations
- unusual activation volumes
- excessive transfers
- credentials without device binding
- abnormal simultaneous sessions
- unusual scan behavior.

Acceptance Condition

Administrators can centrally understand the security posture and configuration of all digital access credentials.

Screen 3.2 — Dynamic QR Security Profile Builder

Purpose

Configure how a dynamic QR is generated and protected.

The matrix requires a unique QR per issued ticket/pass and periodic QR refresh to reduce screenshot and duplication fraud.

QR Configuration

QR Mode

- Static
- Dynamic
- Dynamic + Device Bound
- Dynamic + Location Bound
- Dynamic + Device + Location Bound

Refresh Policy

Configure:

Refresh Every: 30 seconds

Options:

- 15 sec
- 30 sec
- 45 sec
- 60 sec
- Custom

Secure Token Components

Configuration may include:

- Credential ID
- Ticket ID
- Timestamp
- Nonce / OTP
- Device binding reference
- Venue context
- Entitlement payload
- cryptographic signature/key reference.

Important: the UI should configure the **security policy**, while private signing keys/secrets themselves must be protected by the platform's secure key-management layer rather than exposed in ordinary administrator screens.

QR Lifecycle

Visual:

Generate → Display → Refresh → Validate → Rotate → Expire

Acceptance Condition

Administrators can create reusable dynamic-QR security profiles without configuring individual tickets manually.

Screen 3.3 — Credential Activation & Display Rules

Purpose

Configure **when the guest is permitted to see/use the credential**.

The matrix specifies that after registration, a ticket may appear as a blurred QR and only become clear and usable near the park entrance.

Credential States

TICVAI should support:

Registered

↓

Hidden / Blurred

↓

Eligible for Activation

↓

Active

↓

Validated

↓

Expired / Consumed / Revoked

Activation Conditions

Configure:

Visit Date

Must equal ticket visit date

AND

Location

Within authorized geofence

AND/OR

BLE

Authorized beacon detected

AND

Ticket Status

Valid

AND

Device

Registered device

Display Behaviour

Before activation:

- Hide QR
- Blur QR
- Show countdown

- Show "Available at Venue"
- Show venue directions

When active:

- Display dynamic QR
- Display activation timer
- Display credential status
- Display remaining entitlements.

Acceptance Condition

The credential is only displayed in a usable form when its configured activation conditions are satisfied.

Screen 3.4 — Device Binding & Session Security

Purpose

Prevent one credential from being shared across unauthorized devices.

The source explicitly requires tickets to be linked to a specific device/user and suspicious patterns such as device sharing and multiple simultaneous sessions to be detected.

Binding Policy

Configure:

Device Binding

Required

Maximum Active Devices

1

Concurrent Sessions

1

Device Change

- Not allowed
- Allowed before first use
- OTP verification required
- Operator approval required
- Supervisor approval required

Device Record

Show:

- User
- Credential
- Device ID/reference
- App installation
- OS
- Registration date
- Last activation
- Last known venue

- Security status.

Suspicious Behaviour

Detect:

Credential active on Device A

and simultaneously:

Activation attempted on Device B

Response:

BLOCK + SECURITY ALERT

Acceptance Condition

A credential cannot be duplicated across unauthorized devices when device binding is enabled.

Screen 3.5 — BLE Beacon & Geofence Configuration

Purpose

Configure location-aware credential activation.

This is a major requirement under 3.1.9. The matrix requires BLE beacon proximity and geofence boundaries to activate/deactivate credentials at venue, attraction, zone and gate level.

Map-Based Configuration

Show venue map from Board 1.

Administrator selects:

Venue → Park → Zone → Gate

and draws:

Geofence

Example:

Main Entrance Activation Zone — 150 m

BLE Beacon Registry

Configure:

- Beacon Name
- Beacon ID
- Venue
- Zone
- Gate
- Proximity threshold
- Active/Inactive
- Health
- Last detected.

Activation Logic

Example:

IF

Inside Adventure Park Geofence

OR

Beacon MAIN-ENTRY-01 detected

THEN

Activate Dynamic QR

Policy Levels

Support:

- Venue
- Park
- Attraction
- Zone
- Gate.

Acceptance Condition

Digital credentials can automatically activate/deactivate according to configured physical-location context.

Screen 3.6 — Credential Transfer & Rebinding

Purpose

Securely manage digital-ticket transfers.

The matrix requires tickets to be transferable through email/app, with the recipient required to authenticate before accessing and activating the transferred QR.

Transfer Policy

Configure:

- Transfer allowed
- Number of transfers
- Transfer deadline
- Before first validation only
- Require recipient account
- Require OTP
- Require acceptance
- Cancel pending transfer
- Return to sender
- Transfer audit required.

Transfer Journey

Sender Credential

- Transfer Request
- Original QR immediately protected
- Recipient Invitation
- Recipient Login
- Accept
- New Device Binding

→ New Credential Activation

Critical Security Behaviour

After successful transfer:

Old Credential → INVALID

Old Device Binding → REMOVED

Recipient Credential → ACTIVE/ELIGIBLE

Acceptance Condition

Transferred tickets cannot remain simultaneously usable by both sender and recipient.

Screen 3.7 — Credential Revocation & Lifecycle Events

Purpose

Immediately invalidate credentials when the underlying ticket changes.

Requirement 3.1.4 specifically requires dynamic QR invalidation after refunds, cancellations, transfers, exchanges, upgrades or reissues.

Lifecycle Triggers

Configure actions for:

- Refund
- Cancellation
- Transfer
- Exchange
- Upgrade
- Reissue
- Ticket expiration
- Manual invalidation
- Fraud lock
- Account suspension.

Example

Ticket Upgraded

Old credential:

REVOKED

Old QR:

INVALID

New credential:

GENERATED

New entitlement:

SYNCHRONIZED

Revocation Distribution

Show:

Central Platform ✓

Mobile App ✓**Gate Network ✓****Offline Revocation Package ✓****Wallet/Credential Service ✓****Acceptance Condition**

A revoked or replaced credential cannot remain valid within the access ecosystem beyond the configured synchronization/offline security policy.

Screen 3.8 — Offline Cryptographic Validation Profile

Purpose

Allow access devices to validate secure credentials without continuous backend connectivity.

The matrix explicitly requires offline cryptographic validation and embedded entitlement validation without real-time backend connectivity.

Offline Validation Package

Configure what the gate may validate locally:

- Credential authenticity
- Digital signature
- Ticket ID
- Venue
- Park
- Zone
- Visit date
- Time window
- Credential status snapshot
- Ticket type
- Guest category
- Seat
- Timeslot
- Reservation
- Entitlements
- Re-entry permissions
- validity period.

Local Security Package

Devices receive:

Trusted verification material

Rules

Revocation information

Entitlement definitions

Validity parameters

Offline Policy

Example:

Maximum Offline Duration: 8 hours

After threshold:

- Continue restricted validation
- Operator warning
- Supervisor mode
- Fail closed
- configurable fallback.

Reconnection

When connectivity returns:

Upload scans → Resolve state → Update consumption → Detect conflicts → Refresh local package

Acceptance Condition

Supported credentials can be securely validated offline without exposing private signing secrets to gate devices.

Screen 3.9 — Embedded Entitlement Payload Designer

Purpose

Determine what operational information can be securely carried by the credential for offline decisions.

The matrix permits embedded information including ticket type, seat assignment, event ID, venue access rights, timeslot, reservations, locker assignments, membership entitlements, guest category and validity.

Payload Builder

Administrator selects:

Identity

- Credential ID
- Ticket type
- Guest category

Access

- Venue
- Park
- Zone
- Attraction permissions

Validity

- Date
- Time
- Timeslot
- expiry

Entitlements

- Admission

- Fast Pass
- Membership
- Reservation
- Locker
- Seat
- other permitted operational claims.

Payload Preview

TICVAI should show a **human-readable representation**, not cryptographic secrets.

Example:

Venue: Adventure Park

Valid: 29 Aug 2026

Guest: Adult

Park Entry: 1

Fast Pass: 3

Re-entry: 1

Payload Optimization

System estimates:

Payload Size

QR Density

Scan Reliability

Offline Coverage

AI

AI can recommend:

"Locker assignment does not need to be embedded for this credential profile because all locker validation points are online."

Acceptance Condition

Administrators can determine the minimum secure information required for reliable offline admission.

Screen 3.10 — Credential Security Simulation, Audit & Publication

Purpose

Test the complete secure credential lifecycle before production deployment.

Security Simulation

Administrator chooses a scenario:

Valid Credential

Screenshot / Stale QR

Wrong Device

Outside Geofence

No Beacon

Transferred Ticket

Refunded Ticket

Offline Gate**Expired QR****Duplicate Session****Example Test****Credential:**

Annual Pass AP-10452

Device:

Registered

Location:

Main Entrance

QR age:

72 seconds

Configured maximum:

60 seconds

Result**DENIED****Reason:**

DYNAMIC QR EXPIRED

Security Trace**Show:**

- ✓ Signature valid
- ✓ Device valid
- ✓ Venue valid
- ✓ Entitlement valid
- ✗ QR freshness failed

Audit**Every important event records:**

- Credential
- Ticket
- Guest/account reference
- Device
- Gate
- Location
- Timestamp
- Operator
- Activation
- Refresh
- Validation
- transfer
- revocation

- result/reason code.

This aligns with the matrix requirement for complete scan history and entry attempts.

Publication Lifecycle

Draft → Security Test → Validate → Approval → Publish

Support:

Version

Scheduled Activation

Rollback

Emergency Disable

Acceptance Condition

Credential-security configurations can be tested, approved, deployed, audited and rolled back before affecting live access.

Board 3 — Final 10-Screen Structure

#	Backend Screen	Responsibility
3.1	Digital Credential Security Command Center	Overall credential-security configuration and health
3.2	Dynamic QR Security Profile Builder	QR generation, rotation and protection
3.3	Credential Activation & Display Rules	Hidden/blurred/active credential lifecycle
3.4	Device Binding & Session Security	Device and concurrent-session protection
3.5	BLE Beacon & Geofence Configuration	Location-aware activation
3.6	Credential Transfer & Rebinding	Secure ticket transfer
3.7	Credential Revocation & Lifecycle Events	Refund/cancel/exchange/reissue invalidation
3.8	Offline Cryptographic Validation Profile	Secure offline validation
3.9	Embedded Entitlement Payload Designer	Offline credential information
3.10	Security Simulation, Audit & Publication	Test, govern and deploy

Board 3 workflow

Select Credential Type → Create Security Profile → Configure Dynamic QR → Define Activation → Bind Device → Configure Location Rules → Configure Transfer/Revocation → Define Offline Payload → Simulate Security → Approve & Publish

Important boundary with Board 4

There is a deliberate distinction:

Board 3 = Credential security.

It answers: *How is the digital credential protected?*

Board 4 = Media, Credential & Verification Methods.

It answers: *Which physical/digital media and verification methods can represent the ticket, and how are QR, barcode, RFID, NFC, wallet, external credentials and media swapping configured?*

That separation is important because the source matrix requires Access Control to transparently support multiple media technologies while also imposing much deeper security requirements specifically on dynamic digital credentials.

Access Control Module — Board 4

Media, Credential & Verification Methods

Board 4 of 12 — 10 Backend Configuration Screens



Board 4 manages the **credential/media abstraction layer** used by Access Control.

The goal is that an access rule should **not care whether the guest presents a QR code, barcode, RFID wristband, NFC credential, mobile wallet pass, smartphone, hotel room card, biometric identity, or partner-issued ticket**. The system resolves the presented medium to the same virtual ticket/credential and then sends it to the Access Rules Engine from Board 2.

This directly addresses the matrix requirements for transparent validation across barcode, QR, RFID and NFC; media swapping; randomized unique ticket IDs; third-party ticket formats; smartphone reading; multi-range RFID; hotel cards; and hardware-independent scan processing.

Architectural principle

Virtual Credential ID = Master Identity

Multiple media can represent that identity:

Virtual Credential

- QR
- Barcode
- RFID
- NFC
- Mobile Wallet
- Face Pass
- Face Tag
- External/Partner Credential

This is particularly important because changing the physical media should **not create a new ticket or lose the ticket's entitlements, history or access state**.

Screen 4.1 — Media & Credential Command Center

Purpose

Central management page for every media and verification technology supported by Access Control.

KPI Cards

Show:

- Active Media Profiles
- QR Credentials
- RFID Credentials
- NFC Credentials
- Wallet Credentials
- Biometric Credentials
- External Credentials
- Media Swaps Today
- Failed Media Reads
- Unknown Credentials
- Verification Exceptions

Media Directory

Media Profile	Technology	Credential Type	Status	Offline
Mobile Dynamic QR	QR	Digital Ticket	Active	Yes
RFID Wristband	RFID	Wristband	Active	Yes
Annual Pass NFC	NFC	Membership	Active	Yes
Apple Wallet Pass	Wallet	Digital Pass	Active	Yes
Partner QR	QR	External Ticket	Active	Conditional

Filters:

Tenant / Venue / Media / Credential / Product / Status / Integration

AI

AI highlights:

- failing media profiles
- unusual read errors
- duplicate identifiers
- obsolete media
- incompatible device/media combinations.

Acceptance Condition

Administrator has one central view of all access credential and media technologies.

Screen 4.2 — Media Type & Technology Library

Purpose

Define reusable media technologies.

The matrix expects support for linear barcode, two-dimensional codes, magnetic strips, contact/proximity RFID, NFC and biometric readers.

Supported Types

Configure:

Barcode

- 1D / Linear
- 2D
- QR

RFID

- Contact
- Proximity
- ISO 15693
- Other supported standards

NFC

Mobile

- App Credential
- Mobile Wallet

Physical

- Paper Ticket
- Wristband
- Plastic Card
- Hotel Card

Identity

- Face Pass
- Face Tag

External

- Partner QR
- External Barcode
- Third-party Credential.

Properties

Each profile defines:

- technology
- encoding format

- supported reader types
- online/offline capability
- writable/read-only
- security classification
- applicable venues
- applicable products.

Acceptance Condition

New media technologies can be added through configuration/integration rather than changing ticketing business logic.

Screen 4.3 — Virtual Credential & Media Association

Purpose

Associate one virtual ticket identity with its permitted media representations.

Credential View

Example:

Virtual Credential

VC-9837241

Ticket:

Adventure Park Annual Pass

Guest:

Guest 20482

Associated Media

Dynamic QR

Active

RFID Wristband

Active

Mobile Wallet

Provisioned

Face Pass

Enrolled

However, having multiple associated representations does not necessarily mean all may be used simultaneously.

Verification Priority

Administrator configures:

Primary Method

Dynamic QR

Alternatives:

RFID / NFC / Face Pass

Identity Resolution

Regardless of media:

QR 8X72...

or

RFID 298173...

or

Wallet Credential 827...

resolves to:

VC-9837241

and therefore to the same:

- Ticket
- Guest
- Entitlements
- Access history
- Consumption state.

Acceptance Condition

Different media representations resolve consistently to a single virtual credential and access state.

Screen 4.4 — Verification Method Selection & Locking

Purpose

Control which verification method the guest chooses and when it becomes locked.

The matrix specifies that although a ticket may technically support physical card, Dynamic QR, Face Pass and Face Tag, the customer should select **one verification method**. It may be changed before first successful verification, but after that the guest cannot change it; authorized venue operations may do so when necessary.

Configuration

Available Verification Methods

- ☒ Dynamic QR
- ☒ Physical Card
- ☒ RFID
- ☒ Face Pass
- ☒ Face Tag

Selection Policy

Guest selects one

Change Before First Entry

Allowed

Change After First Entry

Guest — Not Allowed

Operations — Supervisor Approval

Example Journey

Ticket Issued

↓

Guest selects:**Dynamic QR**

↓

First successful access

↓

** VERIFICATION METHOD LOCKED****Operator Change****Authorized operator selects:****Change Verification Method****Reason:**

- Lost phone
- Damaged wristband
- Accessibility
- Device failure
- Guest service
- Other.

Every change is audited.

Acceptance Condition

The selected verification method follows configurable locking and authorized-override policies.

Screen 4.5 — Media Issuance & Encoding Profile

Purpose

Configure how ticket identity is written or encoded onto each medium.

The source requires ticket IDs to be generated as 2D barcode, QR or RFID and requires randomized, always-unique identifiers to reduce fraud.

Encoding Profile**Example:****RFID Wristband — Adventure Park****Configure:**

- Credential identifier
- Randomized media identifier
- Ticket ID reference
- Secure token/reference
- Encoding format
- offline payload profile
- checksum/signature where applicable.

Output Profiles**Support:****E-ticket**

M-ticket

Paper Wristband

RFID Wristband

Physical Card

Wallet

Uniqueness Controls

Display:

Identifier Collision Check — ENABLED

Randomization — ENABLED

Duplicate Prevention — ENABLED

Security

TICVAI should never expose sensitive private cryptographic keys on this page.

Instead administrators select a controlled:

Security / Signing Profile

managed by the secure platform layer.

Acceptance Condition

Every issued medium receives a unique and traceable credential identifier using its configured encoding profile.

Screen 4.6 — Media Swap & Replacement

Purpose

Transfer a ticket from one medium to another without changing the underlying virtual ticket.

This directly covers the matrix requirement to swap RFID to barcode/QR or vice versa while transferring the attached information.

Example

Guest loses RFID wristband.

Operator searches:

VC-9837241

Current Media:

RFID-882910

Select:

SWAP MEDIA

From:

RFID Wristband

To:

Dynamic QR

Transfer

Preserve:

- ✓ Ticket
- ✓ Guest
- ✓ Remaining entitlements
- ✓ Entry history
- ✓ Re-entry status
- ✓ Fast Pass balance
- ✓ Reservations
- ✓ Membership association

Old RFID:

REVOKED

New QR:

ACTIVE

Swap Reasons

- Lost
- Damaged
- Device change
- Upgrade
- Guest request
- Operational replacement
- Fraud/security
- Accessibility.

Approval

High-risk swaps may require:

Supervisor Approval

Acceptance Condition

Media can be replaced without changing or duplicating the underlying ticket entitlement.

Screen 4.7 — RFID & NFC Configuration

Purpose

Provide dedicated configuration for proximity credentials.

The matrix requires RFID—including ISO 15693—and NFC, as well as multi-range RFID scanning at near, medium and far ranges.

RFID Profiles

Configure:

- RFID standard
- tag/card type
- frequency/interface profile
- reader compatibility
- read/write behavior
- encoding profile
- security profile.

Reader Range**Near**

Example use:

Individual attraction validation

Medium

Example:

Fast Pass lane

Far

Example:

Seamless guest detection

Range Policy

Administrator can associate range with:

- Venue
- Zone
- Gate
- Credential
- Journey.

NFC**Configure:**

- NFC ticket
- membership
- mobile device
- wallet credential
- supported readers
- offline capability.

Collision Handling

For far-range RFID, configure behavior when multiple credentials are detected:

Identify → Resolve → Validate → Prevent Duplicate Count

Acceptance Condition

RFID/NFC technologies can be configured according to operational range, reader capability and access journey.

Screen 4.8 — External & Partner Credential Mapping**Purpose**

Allow TICVAI Access Control to understand credentials generated by other systems.

The matrix explicitly requires reading reseller/external partner ticket formats and barcodes generated by other systems.

Partner Profile**Example:**

Partner: Hotel Package Provider

Credential:

External QR

Mapping**External:**

ProductCode

→ TICVAI:

Ticket Product

External:

VisitDate

→ TICVAI:

Validity Date

External:

GuestType

→ TICVAI:

Guest Category

External:

Entitlement

→ TICVAI:

Access Entitlement

Validation Mode**Configure:**

- Local Mapping
- API Validation
- Token Validation
- Cached Validation
- Offline Mapping
- Hybrid.

Unknown Credential**If unrecognized:**

DENY

or

Refer to Operator

depending on policy.

Testing**Provide:****Upload Sample Barcode/QR**

→ Decode

→ Map

→ Simulate

→ Validate.

Acceptance Condition

Approved external ticket formats can be recognized and normalized into TICVAI's standard access transaction model.

Screen 4.9 — Hotel, Wallet & External Media Integration

Purpose

Configure specialized external credential ecosystems.

The source specifically requires hotel room-card integration at access points and an interface to the hotel's property-management system for room billing.

Hotel Credential

Configure:

Hotel

PMS Integration

Room Card Provider

Card Format

Eligible Parks

Package Mapping

Example

Guest presents:

Hotel Room Card

TICVAI resolves:

Room 1408

Package:

2 Adults + 2 Children / 3-Day Park Access

Access decision proceeds normally.

Billing Integration

Where supported:

Attraction / Service

→ Guest Room

→ PMS

→ Room Charge

Wallet Credential

Configure supported digital wallet credential profiles and their mapping to the TICVAI virtual credential.

Other External Media

Architecture should support future credential sources without creating separate access-control logic.

Acceptance Condition

Hotel cards, supported wallet credentials and specialized external media can participate in standard TICVAI access journeys.

Screen 4.10 — Media Compatibility, Testing & Publication

Purpose

Ensure that every configured medium works with the intended access-control hardware before deployment. This is particularly important because the matrix says the solution should operate with hardware selected by the venue, while hardware limitations must be highlighted and recommended equipment exposed for procurement decisions.

Compatibility Matrix

Media	Turnstile A	Handheld	VIP Gate	Attraction
Dynamic QR	✓	✓	✓	✓
RFID	✓	✓	✓	✓
NFC	✓	✓	✓	✓
Face Pass	✓	—	✓	✓
Partner QR	✓	✓	✓	✓

Test Console

Select:

Media

RFID Wristband

Reader

Main Gate Reader 03

Credential

Test Credential

Run Test

Results:

- ✓ Media detected
- ✓ Identifier decoded
- ✓ Virtual credential resolved
- ✓ Entitlement loaded
- ✓ Access engine reached
- ✓ Gate response received

Hardware Limitation Detection

Example:

Compatibility Warning

Reader TRN-14 does not support NFC validation in offline mode.

AI Recommendation

Gate 12 receives approximately 2,400 guests/hour. The configured reader profile may create a throughput bottleneck.

Publication

Draft → Compatibility Test → Validate → Approval → Publish

Deployment by:

- Tenant
- Venue
- Park
- Gate
- Device group.

Acceptance Condition

Media profiles cannot be deployed to incompatible access devices without an explicit warning/authorized exception.

Board 4 — Final 10 Screens

#	Backend Screen	Main Responsibility
4.1	Media & Credential Command Center	Overall media estate
4.2	Media Type & Technology Library	Barcode, QR, RFID, NFC, wallet, biometric, etc.
4.3	Virtual Credential & Media Association	Connect multiple media to one ticket identity
4.4	Verification Method Selection & Locking	Guest method selection and first-use lock
4.5	Media Issuance & Encoding Profile	Unique identifier and encoding configuration
4.6	Media Swap & Replacement	RFID ↔ QR ↔ NFC etc.
4.7	RFID & NFC Configuration	Proximity and multi-range credential setup
4.8	External & Partner Credential Mapping	Reseller and third-party ticket recognition
4.9	Hotel, Wallet & External Media Integration	Hotel cards and specialized credentials
4.10	Media Compatibility, Testing & Publication	Hardware compatibility and deployment

Board 4 key workflow

Create Media Type → Configure Technology → Associate Virtual Credential → Define Verification Options → Configure Encoding → Configure RFID/NFC → Map External Credentials → Test Hardware → Approve → Publish

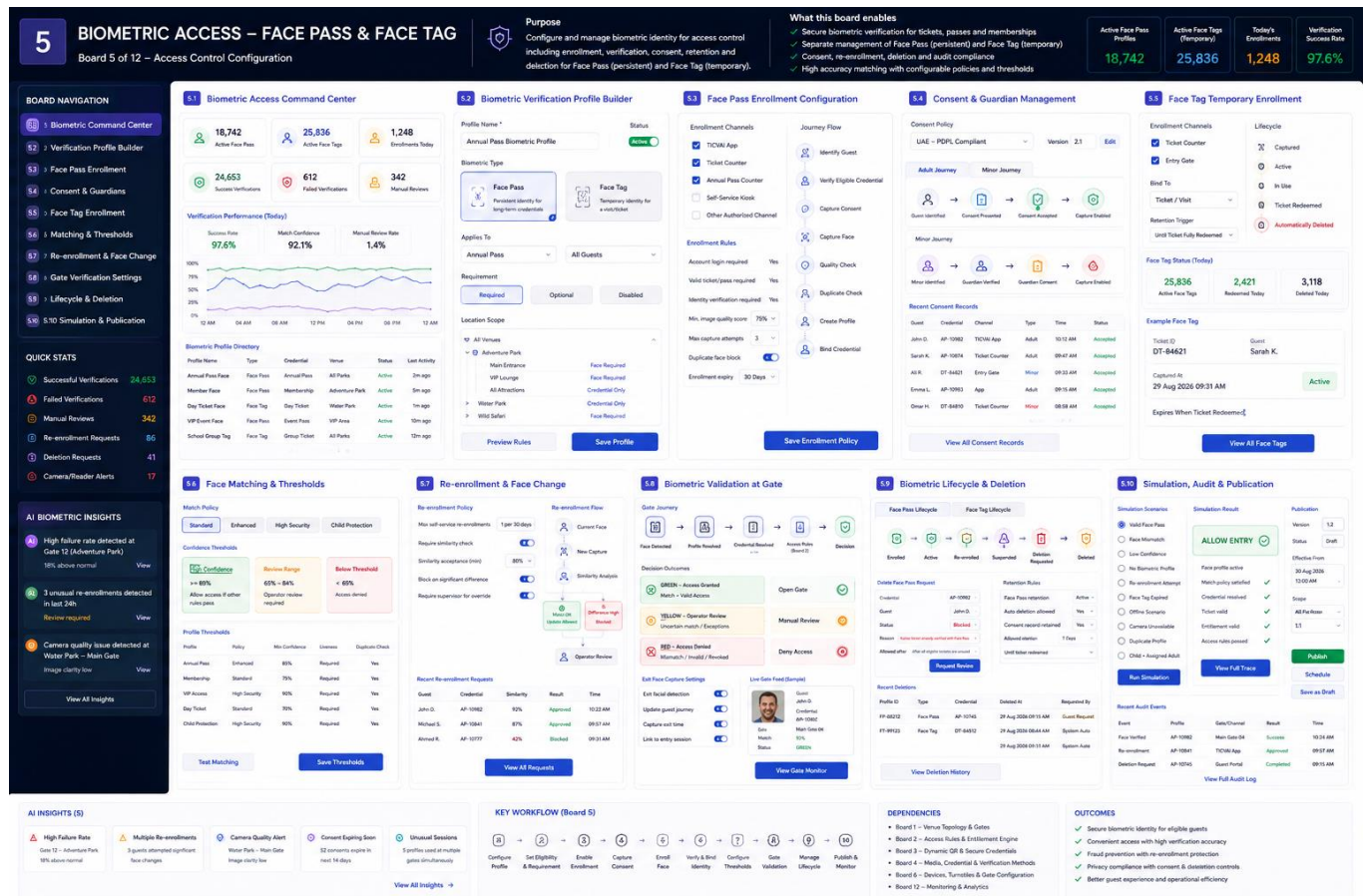
The most important concept for the Board 4 visual design should be the **Virtual Credential** at the center. Instead of showing QR, RFID, NFC and Face Pass as separate tickets, the design should visually communicate:

ONE VIRTUAL TICKET ID → MANY POSSIBLE MEDIA → ONE ACCESS HISTORY → ONE ENTITLEMENT BALANCE

That is the architecture that will make TICVAI genuinely media-independent and prevent the Access Control engine from becoming tied to particular hardware technologies.

Access Control Module — Board 5

Biometric Access — Face Pass & Face Tag



Board 5 of 12 — 10 Backend Configuration Screens

Board 5 manages the **biometric identity and facial-verification layer** of TICVAI Access Control.

The attached matrix makes a clear distinction between two biometric models:

- Face Pass** — persistent biometric identity used for longer-term credentials such as annual passes/memberships, with enrollment, consent, modification and deletion controls.
- Face Tag** — temporary facial identity associated with a ticket/journey and automatically deleted after the ticket is fully redeemed.

Board 5 therefore needs to manage the entire lifecycle:

Eligibility → Consent → Enrollment → Identity Binding → Verification → Exceptions → Re-enrollment → Retention/Deletion → Audit

A major architectural rule should be:

The biometric profile is an identity/verification method—not the ticket itself.

The face resolves to the guest's virtual credential, while Board 2 still makes the final access decision.

Screen 5.1 — Biometric Access Command Center

Purpose

Central dashboard for biometric access configuration and operational health.

KPI Cards

Display:

- Active Face Pass Profiles
- Active Face Tags
- Enrollments Today
- Successful Face Verifications
- Failed Verifications
- Manual Reviews
- Re-enrollment Requests
- Blocked Face Changes
- Profiles Pending Deletion
- Camera/Reader Health
- Biometric Security Alerts

Verification Performance**Show:****Face Pass Success Rate****Face Tag Success Rate****Average Match Confidence****Manual Intervention Rate****Enrollment Failure Rate****Biometric Profile Directory**

Profile	Type	Credential	Venue	Status
Annual Pass Face	Face Pass	Annual Pass	All Parks	Active
Member Face	Face Pass	Membership	Adventure Park	Active
Day Ticket Face	Face Tag	Day Ticket	Water Park	Active

AI**AI identifies:**

- abnormal failure rates
- suspicious re-enrollments
- repeated face changes
- camera-quality issues
- unusual biometric sharing patterns
- gates with high manual intervention.

Acceptance Condition

Authorized administrators can monitor and manage the biometric-access environment from one location.

Screen 5.2 — Biometric Verification Profile Builder

Purpose

Configure **which ticket/credential types can or must use biometric verification.**

The matrix specifically requires biometric checks to be configurable by ticket type, including memberships, annual passes, multi-day and multi-attraction products.

Profile

Example:

Annual Pass Biometric Profile

Biometric Type

- Face Pass
- Face Tag
- Supported future biometric provider

Applies To

Select:

- Ticket Product
- Ticket Type
- Membership
- Annual Pass
- Multi-Day Ticket
- Multi-Attraction Ticket
- VIP Credential
- Accreditation
- selected customer segments.

Requirement

Biometric:

Required / Optional / Disabled

Location Scope

Apply at:

- Venue
- Park
- Zone
- Attraction
- Gate.

Example:

Annual Pass

Main Entry → Face Required

Attractions → Credential Only

VIP Lounge → Face Required

Acceptance Condition

Biometric requirements can be activated or deactivated by product, credential and access location without development.

Screen 5.3 — Face Pass Enrollment Configuration

Purpose

Configure persistent Face Pass registration.

The source specifies that Face Pass may be registered through the **App, ticket counters or Annual Pass counter**.

Enrollment Channels

Administrator enables:

- ☒ TICVAI App
- ☒ Ticket Counter
- ☒ Annual Pass Counter
- ☐ Self-Service Kiosk
- ☐ Other authorized channel

Enrollment Journey

Identify Guest

→

Verify Eligible Credential

→

Capture Required Consent

→

Capture Face

→

Quality Check

→

Duplicate/Identity Check

→

Create Biometric Profile

→

Bind Virtual Credential

Enrollment Rules

Configure:

- Account login required
- Valid ticket/pass required
- Identity check required
- Number of capture attempts
- minimum image quality
- duplicate face detection
- operator verification

- enrollment expiry.

Annual Pass Restriction

The matrix specifically says:

One facial profile may not be associated with more than one annual pass.

The system therefore detects attempts such as:

Face Profile F-8721

already associated with:

Annual Pass AP-10982

Result:

DUPLICATE ASSOCIATION BLOCKED

Acceptance Condition

Face Pass enrollment follows a controlled, configurable and fully auditable workflow.

Screen 5.4 — Biometric Consent & Guardian Management

Purpose

Manage consent requirements associated with persistent biometric enrollment.

The matrix requires App users to provide consent before Face Pass registration and requires guardian consent for minors. On-site enrollment also requires consent before facial data is captured.

Consent Policy

Configure by:

- Country/Jurisdiction
- Tenant
- Venue
- Credential
- Enrollment Channel
- Guest Category.

Adult Journey

Guest Identified

→ Consent Presented

→ Accepted

→ Facial Capture Enabled

Minor Journey

Minor Identified

→ Guardian Required

→ Guardian Identity/Relationship

→ Consent

→ Verification

→ Facial Capture Enabled

Consent Record

Store appropriate consent evidence such as:

- Consent record ID
- Policy/version
- Timestamp
- Channel
- Guardian reference where applicable
- Venue
- Operator where applicable
- withdrawal/deletion status.

Critical Separation**The system should distinguish:****Biometric Data**

from

Consent/Audit Record

because the source explicitly requires biometric data to be permanently removed after deletion while the consent record may be retained.

Acceptance Condition

Face Pass capture cannot proceed when the applicable configured consent requirement has not been satisfied.

Screen 5.5 — Face Tag Temporary Enrollment**Purpose**

Configure the temporary biometric model separately from Face Pass.

The matrix describes Face Tag as temporarily stored facial data, enrollable at ticket counters or entry gates, with biometric data permanently deleted once the associated ticket is fully redeemed.

Face Tag Profile**Configure:****Enrollment Channels**

- Ticket Counter
- Entry Gate

Bind To

- Ticket
- Visit
- Temporary Credential

Retention Trigger

Until Ticket Fully Redeemed

Alternative configurable triggers where permitted:

- End of visit
- Ticket expiration

- Credential cancellation
- operational retention threshold.

Journey

Ticket Presented

→

Face Captured

→

Temporary Face Tag

→

Ticket Linked

→

Access Journey

→

Ticket Fully Redeemed

→

BIOMETRIC DATA DELETED

Status

Face Tags should visibly display:

Temporary

rather than appearing identical to persistent Face Pass records.

Acceptance Condition

Temporary biometric identities are clearly separated from persistent biometric profiles and follow their configured automatic deletion lifecycle.

Screen 5.6 — Face Matching & Verification Thresholds

Purpose

Configure biometric verification behavior.

The source requires a configurable **Biometric Check Level** determining the scoring of biometric comparison.

Match Policy

Example:

Standard Entry

Minimum confidence:

Configured Threshold

Decision Bands

Rather than relying on one binary value, TICVAI should support configurable bands:

HIGH CONFIDENCE

→ Allow if all other access rules pass

REVIEW RANGE

→ Yellow / Operator Verification

BELOW THRESHOLD

→ Deny

Different Profiles**Example:**

Access Type	Biometric Policy
Standard Entry	Standard
Annual Pass	Enhanced
VIP	Enhanced
Restricted Event	High
Child Protection	High

Exact numerical thresholds should be defined according to the selected biometric provider's capabilities and venue policy.

Additional Checks**Configure:**

- Liveness check
- Duplicate-face check
- image quality
- capture timeout
- retry quantity
- mask/obstruction handling
- operator fallback.

Acceptance Condition

Biometric verification sensitivity can be configured by access context and provider capability.

Screen 5.7 — Face Change, Re-enrollment & Identity Protection**Purpose**

Prevent guests from replacing a registered biometric identity with another person's face.

The source explicitly states that customers may re-register Face Pass, but the system must compare the new facial data with the previous profile. If the difference exceeds an acceptable threshold, the update is blocked and venue assistance is required.

Re-enrollment Journey

Current Face

→ New Capture

→ Similarity Analysis

Result A**MATCH WITHIN ACCEPTABLE POLICY**

→ Update permitted

Result B

SIGNIFICANT IDENTITY DIFFERENCE

→

CHANGE BLOCKED

→ Security/Guest Service Review

Operator Review

Show:

- Existing profile reference
- New capture reference
- match result
- credential
- guest
- reason for re-enrollment
- previous changes
- operator
- audit history.

Change Reasons

- Appearance change
- Poor original capture
- Technical issue
- Guest request
- Recovery
- Other.

Identity Lock

Option:

Maximum self-service re-enrollments

Example:

1 per 30 days

Further attempts require authorization.

AI Fraud Signal

AI can flag:

High Risk: This credential has attempted three materially different Face Pass registrations within seven days.

Acceptance Condition

Biometric re-enrollment cannot silently replace the identity attached to a credential.

Screen 5.8 — Biometric Validation at Gate

Purpose

Configure how facial verification interacts with the physical access-control journey.

Gate Journey

Guest Approaches

↓

Camera Detects Face

↓

Face Profile Resolved

↓

Virtual Credential Resolved

↓

Board 2 Access Rules Evaluated

↓

Decision returned.

Decision Outcomes

GREEN

Biometric Match + Valid Access

→ Open Gate

YELLOW

Operator Action Required

Examples:

- uncertain match
- personalized ticket exception
- companion verification
- eligibility requires manual check.

RED

Access Denied

Examples:

- face mismatch
- invalid credential
- revoked profile
- no entitlement.

The matrix already requires gate outcomes using lights, messages, pictograms and sounds, so the biometric result should feed the same unified gate-response mechanism rather than create a separate biometric gate workflow.

Exit Capture

The matrix also describes facial-recognition cameras at park exits to capture guest exit times.

Configure:

Exit Facial Detection

Enabled

Update Guest Journey

Yes

Update Exit Time

Yes

Acceptance Condition

Biometric verification integrates transparently into the normal TICVAI admission and exit decision flow.

Screen 5.9 — Biometric Lifecycle, Retention & Deletion**Purpose**

Manage biometric-data lifecycle and deletion rules.

Face Pass Lifecycle

Enrolled → Active → Re-enrolled → Suspended → Deletion Requested → Deleted

Face Tag Lifecycle

Captured → Active → Used → Ticket Redeemed → Automatically Deleted

Face Pass Delete Request

The source specifies that Face Pass deletion should not be allowed when an active ticket has already been verified with Face Pass. Once deletion is permitted, unused valid tickets should move to an alternative verification method.

Example**Guest requests:**

DELETE FACE PASS

System checks:

Active Credential?

YES

Already validated using Face Pass?

YES

Result:

DELETION CURRENTLY BLOCKED

Reason shown.

If eligible:**Delete biometric profile**

- Remove biometric association
- Determine alternative verification method
- Preserve permitted audit/consent record
- Record deletion event.

Automated Retention Rules**Configure separately for:**

- Face Pass
- Face Tag
- Failed enrollment captures
- abandoned registrations
- temporary captures.

Acceptance Condition

Biometric information follows explicit, auditable lifecycle and deletion rules independently from ticket history.

Screen 5.10 — Biometric Simulation, Audit & Publication

Purpose

Test biometric configurations before live deployment.

Simulation Scenarios

Support:

- Valid Face Pass
- Face mismatch
- No biometric profile
- Low-confidence match
- Liveness failure
- Duplicate profile
- Re-enrollment attempt
- Child + assigned adult
- Face Tag expired/deleted
- Offline biometric scenario
- Camera unavailable
- alternative verification fallback.

Example Simulation

Credential

Annual Pass AP-10742

Verification

Face Pass

Location

Main Entrance Gate 04

Biometric

Match Passed

Ticket

Valid

Entitlement

Available

Result**ALLOW ENTRY****Decision Trace:**

- ✓ Face profile active
- ✓ Verification policy satisfied
- ✓ Match policy satisfied
- ✓ Credential resolved
- ✓ Ticket valid
- ✓ Access entitlement valid

Audit Explorer**Search by:**

- Guest
- Credential
- Face profile reference
- Gate
- Device
- Operator
- Date
- Result
- reason code.

Publication

Draft → Test → Privacy/Policy Validation → Approval → Publish

Deployment scope:

- Tenant
- Venue
- Park
- Credential type
- Gate group.

Support:**Versioning****Scheduled activation****Rollback****Emergency disable biometric verification**

with appropriate fallback rules.

Acceptance Condition

No biometric access configuration enters production without testing, authorization, version control and auditability.

Board 5 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
5.1	Biometric Access Command Center	Biometric estate and operational health
5.2	Biometric Verification Profile Builder	Determine where/when biometrics are required
5.3	Face Pass Enrollment Configuration	Persistent biometric enrollment
5.4	Biometric Consent & Guardian Management	Consent and minor/guardian workflow
5.5	Face Tag Temporary Enrollment	Temporary visit-level biometric identity
5.6	Face Matching & Verification Thresholds	Match policy and verification behavior
5.7	Face Change, Re-enrollment & Identity Protection	Prevent biometric identity substitution
5.8	Biometric Validation at Gate	Gate/exit biometric journey
5.9	Biometric Lifecycle, Retention & Deletion	Retention and deletion management
5.10	Biometric Simulation, Audit & Publication	Test, govern and deploy

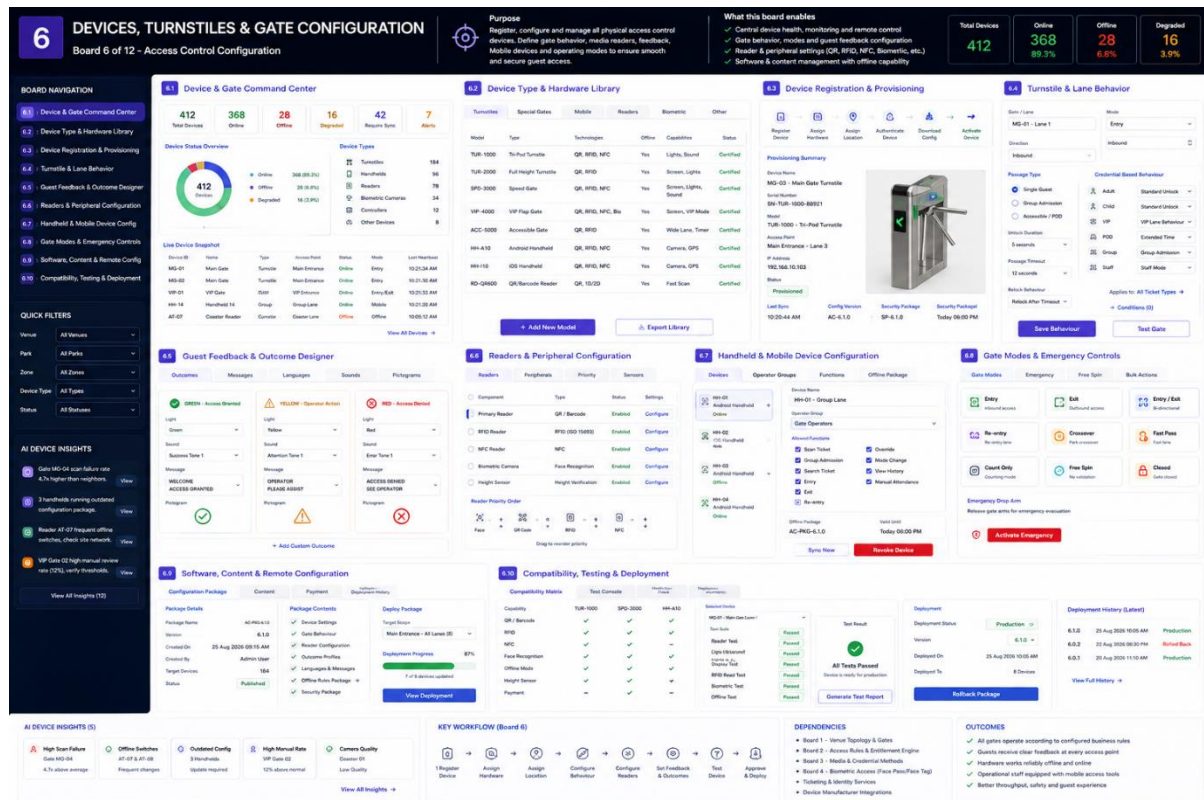
Board 5 key workflow

Select Credential → Choose Face Pass/Face Tag → Configure Eligibility → Configure Enrollment → Capture Required Consent → Bind Identity → Configure Match Policy → Configure Gate Behaviour → Define Retention/Deletion → Simulate → Approve → Publish

One important point for the eventual **Board 5 design**: Face Pass and Face Tag should be visually differentiated throughout the UI. **Face Pass should look like a persistent identity profile**, while **Face Tag should clearly show a temporary lifecycle/countdown/deletion trigger**. This will make a potentially complex biometric configuration much easier for venue administrators to understand.

Access Control Module — Board 6

Devices, Turnstiles & Gate Configuration



Board 6 of 12 — 10 Backend Configuration Screens

Board 6 is the **physical access-control hardware and edge-device configuration layer**.

Boards 1–5 defined the topology, rules, secure credentials, media, and biometrics. Board 6 connects those configurations to the **actual equipment operating at the venue**: turnstiles, accessible gates, VIP gates, Fast Pass lanes, attraction gates, handhelds, smartphones/tablets, QR/barcode readers, RFID/NFC readers, biometric cameras, and other supported controllers.

The source matrix specifically requires support for turnstiles, disabled-access gates, handhelds, VIP gates, Fast Pass lanes, attraction entrances, Android/iOS devices, multiple turnstile sizes, configurable gate behavior, lights/sounds/pictograms, Free Spin, Drop Arm, and hardware selected by the venue.

The architectural principle is:

Hardware should execute TICVAI configuration — business rules should not be hard-coded into the hardware.

Screen 6.1 — Device & Gate Command Center

Purpose

Provide the central operational/configuration view of the complete access-control hardware estate.

KPI Cards

Display:

- Total Devices
- Online
- Offline

- Degraded
- Turnstiles
- Handhelds
- Biometric Readers
- RFID/NFC Readers
- Gates Open
- Gates Closed
- Devices Requiring Sync
- Firmware/Software Exceptions
- Hardware Alerts

Device Directory

Device	Type	Access Point	Mode	Connectivity	Status
MG-01	Turnstile	Main Gate	Entry	Online	Healthy
MG-02	Turnstile	Main Gate	Entry	Online	Healthy
VIP-01	VIP Gate	VIP Entrance	Entry/Exit	Online	Healthy
HH-14	Handheld	Group Lane	Mobile	Wi-Fi	Active
AT-07	Reader	Coaster	Fast Pass	Offline	Local Mode

Live Health

Show:

- connectivity
- last heartbeat
- configuration version
- local rule version
- credential/security package version
- scanner health
- controller health
- camera health where applicable.

AI

AI identifies:

Gate MG-04 has a scan-failure rate 4.7× higher than neighboring gates.

or:

Three handhelds are running an outdated configuration package.

Acceptance Condition

Operations can identify every access-control device and its current operational/configuration state from one screen.

Screen 6.2 — Device Type & Hardware Library

Purpose

Create reusable hardware definitions independently from physical deployed devices.

Device Categories

Support:

Turnstiles

- Standard
- Full Height
- Tripod
- Speed Gate
- Wide Lane

Special Gates

- Accessible/POD Gate
- Buggy Gate
- VIP Gate
- Staff Gate

Mobile

- Android Handheld
- iOS Device
- Tablet

Readers

- QR/Barcode
- RFID
- NFC
- Multi-technology
- Biometric

Other

- Podium
- Counter
- Beacon
- camera/controller
- supported external access device.

The matrix requires operation across handheld/laser/image readers, Android/iOS apps and full-height or standard turnstiles.

Hardware Profile

Each model contains:

- Manufacturer
- Model
- Device Category

- Supported technologies
- Connectivity
- Offline capability
- Screen capability
- Sound capability
- Light capability
- Relay/controller support
- payment capability where available
- firmware/software information.

Certification

Status:

Certified

Compatible

Conditional

Unsupported

Acceptance Condition

TICVAI maintains a reusable catalog of supported access-control hardware without embedding manufacturer-specific logic into admission policies.

Screen 6.3 — Physical Device Registration & Provisioning

Purpose

Register actual deployed hardware and connect it to the Board 1 topology.

Device Identity

Configure:

- Device ID
- Serial Number
- Hardware Model
- Manufacturer
- Tenant
- Venue
- Park
- Zone
- Access Point
- Gate/Lane
- IP/network reference
- controller reference
- installation date.

Provisioning

Journey:

Register Device

- Assign Hardware Profile
- Assign Location
- Authenticate Device
- Download Configuration
- Download Security Package
- Connectivity Test
- Activate

Device Authentication

Use controlled device credentials/certificates.

Status:

Provisioned

Pending

Suspended

Revoked

Replacement

Support:

Replace Device

while preserving:

- Gate assignment
- **Configuration**
- Access rules
- Media profiles
- operating mode.

Acceptance Condition

Every physical access-control device is uniquely registered, authenticated and assigned to its correct topology location.

Screen 6.4 — Turnstile & Lane Behavior Configuration

Purpose

Configure how each turnstile or lane behaves.

The matrix specifically requires software on turnstiles to behave differently according to card/ticket type and supports different operating modes.

Gate Mode**Configure:**

- Entry
- Exit
- Entry/Exit
- Re-entry

- Crossover
- Fast Pass
- Attraction
- Group
- Count Only
- Free Spin
- Closed
- Emergency.

Direction

Inbound

Outbound

Bidirectional

Unlock Behaviour

Configure:

Single Guest

1 rotation/open cycle

Group Ticket

N admissions

Accessible Gate

Extended opening duration

Buggy Gate

Wide opening

Credential-Based Behavior

Example:

Standard Adult

→ Normal unlock

Child

→ Yellow indicator + normal unlock

POD

→ Yellow + operator assistance

VIP

→ VIP gate workflow

Group

→ Group admission mode.

Timeout

Configure:

- Unlock duration
- pass-through timeout
- relock behavior

- incomplete passage behavior.

Acceptance Condition

Each gate can execute different configured behaviors according to its mode and the validated credential.

Screen 6.5 — Validation Outcome & Guest Feedback Designer

Purpose

Configure what the physical access device does and displays after validation.

The matrix explicitly requires valid/non-valid messages, lights, pictograms and sounds, including green/yellow/red behavior.

Outcome Profiles

GREEN — ACCESS GRANTED

Configure:

- Green light
- Gate open
- Success tone
- ✓ pictogram
- Custom message

Example:

WELCOME — ACCESS GRANTED

YELLOW — OPERATOR ACTION

Configure:

- Yellow light
- Alert sound
- Gate remains controlled
- operator prompt.

Examples:

CHECK ID

VERIFY COMPANION

MANUAL ELIGIBILITY CHECK

RED — ACCESS DENIED

Configure:

- Red light
- Denial sound
- Gate remains locked
- reason code.

Example:

TICKET EXPIRED

Outcome Rules

Different responses can be configured for:

- Adult
- Child
- VIP
- POD
- Membership
- invalid credential
- wrong verification method
- biometric review
- re-entry exception.

Multilingual Messages

Support:

Arabic / English / other venue languages

Acceptance Condition

Every access decision produces a configurable and understandable physical response for the guest and operator.

Screen 6.6 — Reader, Scanner & Peripheral Configuration

Purpose

Configure the technologies attached to a gate/device.

Components

A gate may contain:

QR Reader

1D/2D Barcode Reader

RFID Reader

NFC Reader

Biometric Camera

Height Sensor

Display

Speaker

Light Tower

Printer

Payment Reader

where supported.

Reader Priority

Example:

1. Face Pass
2. Dynamic QR
3. RFID
4. NFC

or:

Auto Detect

RFID Range

Configure:

Near / Medium / Far

consistent with the multi-range RFID requirements.

Sensor Conditions

Example:

Junior ticket:

Height $\leq$ configured threshold

If supported hardware includes a height sensor:

Automatically Verify

Otherwise:

Yellow → Operator Verification

Device Capability Detection

TICVAI shows:

Height verification is configured, but Gate 14 does not have a supported height sensor.

Acceptance Condition

Readers and peripherals can be independently associated with gates and checked against required admission capabilities.

Screen 6.7 — Handheld & Mobile Access Device Configuration

Purpose

Configure mobile access-control devices used by staff.

The matrix explicitly requires handheld devices and Android/iOS dedicated mobile applications.

Handheld Profile

Configure:

- Device type
- Android/iOS
- assigned venue
- assigned zone
- assigned operator group
- permitted operating modes
- offline capability
- scanner source
- biometric capability where supported.

Operator Functions

Enable/disable:

- Scan Ticket

- Search Ticket
- Entry
- Exit
- Re-entry
- Crossover
- Group Admission
- Manual Attendance
- Override
- View History
- Change Device Mode.

Restricted Functions

Example:

Standard Attendant:

Scan + Group Admission

Supervisor:

Scan + Override + Mode Change

Offline Package

Display:

Last Sync

09:42

Rules Valid Until

18:00

Revocation Package

Current

Lost Device

Action:

REMOTE REVOKE DEVICE

prevents further trusted access transactions.

Acceptance Condition

Mobile access devices can be centrally configured, restricted and revoked according to their operational role.

Screen 6.8 — Gate Modes, Free Spin & Emergency Controls

Purpose

Manage non-standard operational modes.

The source specifically requires **Free Spin** and **Drop Arm/Emergency** behavior.

Normal Modes

- Entry
- Exit

- Re-entry
- Crossover
- Closed.

Free Spin

When activated:

Credential Reading: OFF

Turnstile Rotation Count: ON

Use cases:

- Free View Day
- Open event
- non-ticketed attendance.

System continues counting physical passage.

Count Only

Gate does not make an admission decision but counts movements.

Emergency / Drop Arm

When activated:

Validation OFF

Counting OFF

Gate RELEASED

Emergency Controls

Configure:

- Who can activate
- Venue scope
- Gate group
- reason
- emergency code
- automatic notification
- incident record.

Bulk Command

Example:

SELECT: MAIN ENTRANCE — 18 GATES

→

ACTIVATE EMERGENCY MODE

Safety

High-risk commands should require explicit authorization and create immutable audit records.

Acceptance Condition

Authorized operations personnel can rapidly change gate modes across individual devices or device groups while maintaining full auditability.

Screen 6.9 — Device Software, Content & Remote Configuration

Purpose

Centrally control access-control device software and guest-facing configuration.

The source requires the ability to configure/manage software on turnstiles, add external webpages on supported screens, and enable payment technologies where available.

Configuration Package

Contains:

- Device settings
- Gate mode
- Reader settings
- Media profiles
- Outcome profiles
- Language
- UI content
- Local rules
- Offline security configuration.

Screen Content

Where device hardware supports a screen:

Configure:

- Welcome page
- Instructions
- Ticket status
- Reason message
- Promotional information
- External approved webpage
- Emergency information.

Payment

Where hardware supports payment:

Payment Capability

Enabled / Disabled

Example use cases:

- entitlement upgrade
- Fast Pass purchase
- parking
- supplemental access.

Actual payment processing remains connected to the TICVAI payment layer rather than being implemented independently by the gate.

Remote Deployment

Draft → Test Device → Device Group → Venue Rollout

Show deployment progress:

78 / 80 devices updated

Failures:

2 devices offline

Acceptance Condition

Device configuration and supported content can be centrally deployed without manually configuring each turnstile.

Screen 6.10 — Hardware Compatibility, Health, Testing & Deployment**Purpose**

Provide the final testing and governance layer before hardware is used in production.

The matrix says venues may select their hardware, while the provider must expose hardware limitations and recommendations.

Compatibility Matrix

Capability	Gate A	Gate B	HH-01	VIP-01
Dynamic QR	✓	✓	✓	✓
RFID	✓	✓	✓	✓
NFC	✓	—	✓	✓
Face Pass	✓	—	✓	✓
Offline	✓	✓	✓	✓
Height Check	✓	—	—	✓

Hardware Test**Run:****Reader Test****Gate Open/Close****Light Test****Sound Test****Display Test****RFID Test****NFC Test****Biometric Test****Offline Test****Emergency Mode Test****End-to-End Test****Use test credential:**

Credential Read

- Virtual Credential Resolved
- Access Rule Evaluated
- Entitlement Checked
- Decision Returned
- Light/Sound Triggered
- Gate Opened
- Attendance Updated
- Audit Created.

AI Hardware Advisor**AI can identify:**

Throughput Risk: Gate MG-06 averages 1.8 seconds per QR validation versus the 0.9-second target.

or:

Compatibility Gap: Four attraction gates cannot execute the configured Face Pass policy.

Deployment**Lifecycle:**

Registered → Configured → Tested → Approved → Production

Support:

- Pilot deployment
- Selected gates
- device group
- venue
- scheduled rollout
- rollback.

Acceptance Condition

No access device enters production until its required capabilities, connectivity, security configuration and physical responses have been successfully validated.

Board 6 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
6.1	Device & Gate Command Center	Hardware estate and live health
6.2	Device Type & Hardware Library	Reusable hardware/model definitions
6.3	Physical Device Registration & Provisioning	Register and authenticate deployed devices
6.4	Turnstile & Lane Behavior Configuration	Gate modes, direction and physical behavior

#	Backend Screen	Main Responsibility
6.5	Validation Outcome & Guest Feedback Designer	Green/yellow/red, messages, lights and sounds
6.6	Reader, Scanner & Peripheral Configuration	QR, RFID, NFC, biometrics and sensors
6.7	Handheld & Mobile Access Device Configuration	Android/iOS and staff mobile access
6.8	Gate Modes, Free Spin & Emergency Controls	Free Spin, count-only, closed and Drop Arm
6.9	Device Software, Content & Remote Configuration	Central software/content deployment
6.10	Hardware Compatibility, Health, Testing & Deployment	Certification, testing and production rollout

Board 6 workflow

Select Hardware → Register Device → Assign Gate → Attach Readers → Configure Gate Behavior → Configure Guest Feedback → Configure Mobile/Operator Functions → Configure Special Modes → Push Device Package → Test → Approve → Deploy

Important separation for the design

The Board 6 visual should make the architecture very clear:

Board 2 — Access Rules

Should this guest enter?

↓

Boards 3–5 — Credential / Media / Identity

Who or what is being presented?

↓

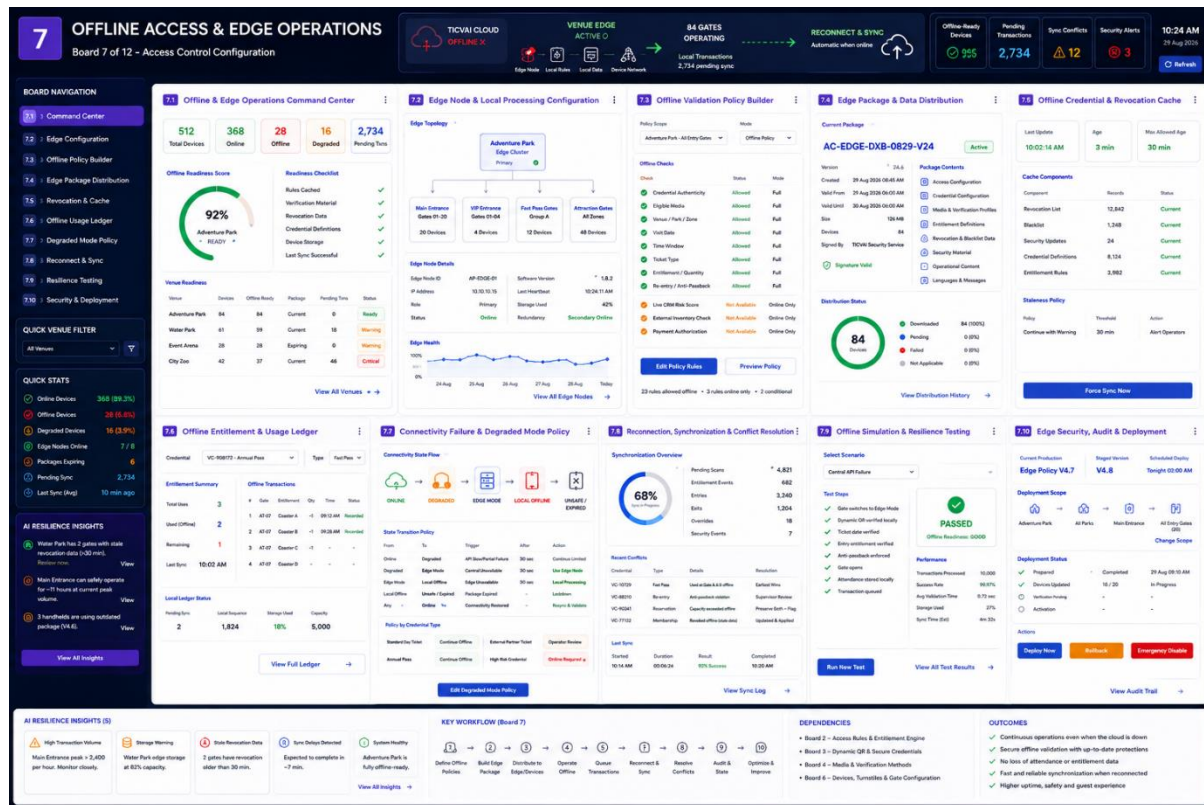
BOARD 6 — PHYSICAL EXECUTION

Reader → Controller → Light / Screen / Sound → Gate → Attendance Event

This means Board 6 should look more like a **modern IoT/device-management control center** than a normal ticketing administration page. It should have live device cards, a graphical gate layout, health indicators, hardware capability matrices, remote controls, configuration deployment status, and AI-detected device issues.

Access Control Module — Board 7

Offline Access & Edge Operations



Board 7 of 12 — 10 Backend Configuration Screens

Board 7 is the **business-continuity and edge-computing layer** of TICVAI Access Control.

The matrix explicitly requires access control to continue operating when the central database/platform is unavailable. Turnstiles and devices must locally validate eligible media, site, date/time, access rights, re-entry/crossover rules, and embedded entitlements, then synchronize transactions when connectivity returns.

This board should therefore answer one critical question:

What happens at every gate if TICVAI Cloud, venue WAN, database, API, or internet connectivity disappears?

The objective is **not simply "offline scanning."** TICVAI should provide a managed **Edge Access Engine** capable of securely executing an approved subset of the central access configuration locally.

Architecture

TICVAI Central Platform

↓ Publish

Signed Edge Package

↓ Distribute

Venue Edge / Gate / Handheld

↓ Local validation

Scan Transactions & State Changes

↑ Synchronize

TICVAI Central Platform

Screen 7.1 — Offline & Edge Operations Command Center

Purpose

Provide a real-time overview of offline readiness across the entire access-control estate.

KPI Cards

Display:

- Offline-Ready Devices
- Currently Online
- Currently Offline
- Devices in Degraded Mode
- Edge Nodes Online
- Packages Current
- Packages Expiring
- Pending Offline Transactions
- Sync Conflicts
- Offline Security Alerts

Venue Readiness

Venue	Devices	Offline Ready	Package	Pending Txns	Status
Adventure Park	84	84	Current	0	Ready
Water Park	61	59	Current	18	Warning
Event Arena	28	28	Expiring	0	Warning

Offline Readiness Score

Example:

Adventure Park — 98% READY

Checks:

- ✓ Rules cached
- ✓ Verification material current
- ✓ Revocation data current
- ✓ Credential definitions available
- ✓ Device storage healthy
- ✓ Last synchronization successful

AI Assistant

AI highlights:

Risk: Two Water Park gates have not synchronized their revocation package for 6 hours.

or:

Capacity Risk: Gate cluster North Entrance has only 43 minutes of estimated offline battery/runtime remaining.

Acceptance Condition

Operations can immediately determine whether each venue, gate and device is capable of safely operating offline.

Screen 7.2 — Edge Node & Local Processing Configuration

Purpose

Configure where local access decisions are processed when central services cannot be reached.

Edge Architecture

Support:

Venue Edge Node

Shared local processing for a venue.

Gate Controller

Local processing at gate level.

Turnstile Local Engine

Embedded device processing.

Handheld Local Engine

Mobile offline processing.

Edge Node Configuration

Fields:

- Edge Node ID
- Tenant
- Venue
- Network
- Device Group
- Processing Mode
- Storage allocation
- redundancy
- last heartbeat
- software version
- security status.

Device Assignment

Visual topology:

Adventure Park Edge Cluster

- Main Entrance Gates 01–20
- VIP Gates 01–04
- Fast Pass Gates
- Attraction Gates
- Handheld Group A

High Availability

Where infrastructure supports it:

Primary Edge Node

↓

failure

↓

Secondary Edge Node**Acceptance Condition**

Administrators can define which edge component is responsible for local access decisions for every deployed device.

Screen 7.3 — Offline Validation Policy Builder**Purpose**

Define exactly which access checks are allowed to run locally.

The matrix identifies key offline criteria including eligible media, eligible site, eligible time, ticket validity and eligible access mode.

Offline Checks**Administrator selects:**

- ☒ Credential Authenticity
- ☒ Eligible Media
- ☒ Venue
- ☒ Park
- ☒ Zone
- ☒ Attraction
- ☒ Visit Date
- ☒ Time Window
- ☒ Ticket Type
- ☒ Guest Category
- ☒ Entitlement
- ☒ Entry Quantity
- ☒ Re-entry
- ☒ Crossover
- ☒ Anti-Passback
- ☒ Fast Pass
- ☒ Membership
- ☒ Blacklist/Revocation Snapshot

Rule Classification

Every Board 2 rule should automatically show:

OFFLINE COMPATIBLE

or

ONLINE REQUIRED

or

CONDITIONAL**Example****Rule:**

Ticket Date = Today

Offline:

SUPPORTED**Rule:**

Live CRM Risk Score < 70

Offline:**NOT AVAILABLE****Fallback:**

Use Cached Risk Score

Acceptance Condition

Administrators explicitly control which admission decisions may be performed without central connectivity.

Screen 7.4 — Edge Package & Data Distribution

Purpose

Define what configuration and operational data is securely distributed to edge devices.

Edge Package Contents**Package can contain:****Access Configuration**

- venues
- zones
- gates
- access rules
- calendars

Credential Configuration

- media profiles
- verification profiles
- entitlement definitions

Security

- trusted verification material
- revocation information
- credential security parameters

Operational

- reason codes
- gate responses
- languages
- operator permissions.

Package Example

AC-EDGE-DXB-0829-V24

Version:

24.6

Devices:

84

Size:

126 MB

Valid:

29 Aug – 30 Aug

Distribution**Central Platform**

→ Venue Edge

→ Device Groups

→ Individual Devices

Package Status**Each device displays:****Downloaded****Verified****Activated****Failed****Integrity Check****Before activation:**

- ✓ Signature valid
- ✓ Package complete
- ✓ Version valid
- ✓ Device authorized

Acceptance Condition

Only approved, intact and authorized edge packages can become active on access-control devices.

Screen 7.5 — Offline Credential & Revocation Cache

Purpose

Manage the local information required to reject credentials that should no longer be usable.

This is especially important because Board 3 requires refunded, cancelled, transferred, exchanged, upgraded and reissued credentials to be invalidated.

Cache Configuration**Configure:****Credential Cache****Revocation List****Blacklist****Security Updates****Entitlement Snapshot****Priority Updates****Critical events:**

- Fraud Lock
- Refund
- Cancellation
- Lost Credential
- Transfer
- Reissue
- Manual Invalidation

can trigger:

URGENT EDGE DISTRIBUTION

Cache Age

Display:

Last Updated

10:02:14

Age

3 min

Maximum Allowed

30 min

Staleness Policy

If revocation cache exceeds allowed age:

- Continue
- Continue with warning
- Restricted products only
- Supervisor mode
- Deny selected credential classes
- Fail closed.

Acceptance Condition

Offline devices have a controlled and measurable mechanism for receiving credential invalidations and security changes.

Screen 7.6 — Offline Entitlement & Usage Ledger

Purpose

Track entitlement consumption while the central system is unavailable.

This is necessary for tickets such as:

3 Fast Pass uses

1 park entry

1 meal

1 re-entry

where usage can occur during an outage.

Local Ledger

Example:**Credential:**

VC-908172

Before outage:

Fast Pass Remaining: 3

Offline transaction:

Ride A → -1

Local balance:

2

Ride B → -1

Local balance:

1

Offline Transaction Record**Store:**

- Credential
- Device
- Gate
- Entitlement
- Quantity
- Timestamp
- local sequence
- operator
- decision
- package version.

Duplicate Prevention**Where possible:****Local Device**

and

Venue Edge

share current usage state.

Risk Policy**For high-risk entitlements, administrator may configure:**

Online Only

or:

Offline Allowed — Maximum 1

Acceptance Condition

Offline usage is recorded locally and prevents repeated consumption within the available edge synchronization scope.

Screen 7.7 — Connectivity Failure & Degraded Mode Policy

Purpose

Configure how devices transition from normal online operation to offline/degraded operation.

Connectivity State

TICVAI should distinguish:

ONLINE

Central services reachable.

DEGRADED

Some services unavailable.

EDGE MODE

Central unavailable but venue edge available.

LOCAL OFFLINE

Only device-local processing available.

UNSAFE / EXPIRED

Required offline package no longer trusted.

Transition Rules

Example:

Central unavailable:

30 sec

↓

Switch to:

EDGE MODE

Edge unavailable:

30 sec

↓

Switch to:

LOCAL OFFLINE

Guest Experience

The switch should normally happen automatically without stopping guest flow.

Device Display

Operator sees:

OFFLINE MODE ACTIVE

Package:

Valid

Last Sync:

10:04

Policy by Credential

Example:

Standard Day Ticket:

Continue Offline

Annual Pass:

Continue Offline

External Partner Ticket:

Operator Review

High-Risk Credential:

Online Required

Acceptance Condition

Connectivity failures trigger predefined operating modes rather than unpredictable gate behavior.

Screen 7.8 — Reconnection, Synchronization & Conflict Resolution

Purpose

Synchronize everything that occurred offline when connectivity returns.

Reconnection Workflow

Connectivity Restored

↓

Authenticate

↓

Upload Offline Transactions

↓

Sequence Events

↓

Reconcile Credential State

↓

Reconcile Entitlements

↓

Update Attendance

↓

Resolve Conflicts

↓

Download Latest Configuration

↓

Return Online.

Sync Queue

Show:

Pending Scans: 4,821

Entitlement Events: 682

Entries: 3,240

Exits: 1,204

Overrides: 18

Security Events: 7

Conflict Example

Credential:

VC-10729

Gate A offline:

Fast Pass used

Gate B offline:

Fast Pass used

Central remaining balance before outage:

1

Result:

CONFLICT

Resolution Policies

- Preserve both + flag
- earliest transaction wins
- configured business rule
- supervisor review
- security investigation.

Never silently discard an access transaction.

Acceptance Condition

Offline events synchronize back to the central platform with deterministic reconciliation and a complete audit trail.

Screen 7.9 — Offline Simulation & Resilience Testing

Purpose

Allow venues to prove that their access environment will survive outages **before opening to guests**.

Test Scenarios

Select:

Central API Failure

Database Unavailable

Internet Failure

Venue WAN Failure

Edge Node Failure

Device Isolated

Revocation Cache Stale

Expired Edge Package

High Offline Volume

Reconnect After Extended Outage

Test Journey**Example:****SIMULATE CENTRAL OUTAGE****TICVAI checks:**

- ✓ Gate switches to Edge Mode
- ✓ Dynamic QR verified locally
- ✓ Ticket date verified
- ✓ Entry entitlement verified
- ✓ Anti-passback enforced
- ✓ Gate opens
- ✓ Attendance stored locally
- ✓ Transaction queued

Result**OFFLINE READINESS: PASSED****Load Test****Example:****10,000 validations****during:****2-hour outage****Measure:**

- validation speed
- storage
- synchronization time
- error rate.

AI Resilience Advisor

Main Entrance can safely operate for approximately 11 hours at projected peak transaction volume before local transaction storage reaches the configured warning threshold.

Acceptance Condition

Offline configuration can be tested under realistic outage scenarios before production use.

Screen 7.10 — Edge Security, Audit & Deployment**Purpose**

Govern the complete offline/edge environment.

Security Dashboard**Show:**

- Edge certificates/credentials
- Package signatures
- Authorized devices
- Revoked devices
- Failed package validation
- unauthorized connection attempts

- configuration changes
- offline override activity.

Deployment Lifecycle

Draft

- Validate
- Security Test
- Offline Simulation
- Approval
- Pilot
- Publish

Deployment Scope

- Tenant
- Venue
- Park
- Edge Cluster
- Gate Group
- Device Group
- Individual Device.

Version Control

Example:

Edge Policy V4.8

Current Production:

V4.7

Scheduled:

V4.8 — Tonight 02:00

Rollback

ROLLBACK TO V4.7

must restore:

- Offline rules
- Device configuration
- Security policy
- edge package definitions.

Audit

Record:

Who changed what

When

Previous value

New value

Approval

Deployment target**Result****Emergency Action**

Authorized security administrators can:

REVOKE EDGE PACKAGE

DISABLE DEVICE

FORCE RESYNC

FORCE ONLINE-ONLY

where operationally appropriate.

Acceptance Condition

Offline and edge configurations are version-controlled, secured, approved, deployable and fully auditable.

Board 7 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
7.1	Offline & Edge Operations Command Center	Estate-wide offline readiness
7.2	Edge Node & Local Processing Configuration	Venue/device edge architecture
7.3	Offline Validation Policy Builder	Determine which rules work offline
7.4	Edge Package & Data Distribution	Secure local configuration packages
7.5	Offline Credential & Revocation Cache	Local invalidation/security state
7.6	Offline Entitlement & Usage Ledger	Track consumption during outages
7.7	Connectivity Failure & Degraded Mode Policy	Automatic fallback behavior
7.8	Reconnection, Synchronization & Conflict Resolution	Reconcile offline transactions
7.9	Offline Simulation & Resilience Testing	Test outage scenarios
7.10	Edge Security, Audit & Deployment	Secure and govern edge operations

Board 7 key workflow

Define Edge Architecture → Select Offline Rules → Build Edge Package → Configure Revocation Cache → Configure Offline Entitlements → Define Degraded Modes → Configure Reconciliation → Simulate Outage → Security Validate → Approve & Deploy

The Board 7 design should visually communicate something very important: **the venue does not stop when the cloud goes down.**

I would make the central visual on the board show:

TICVAI CLOUD — OFFLINE X



VENUE EDGE — ACTIVE ✓



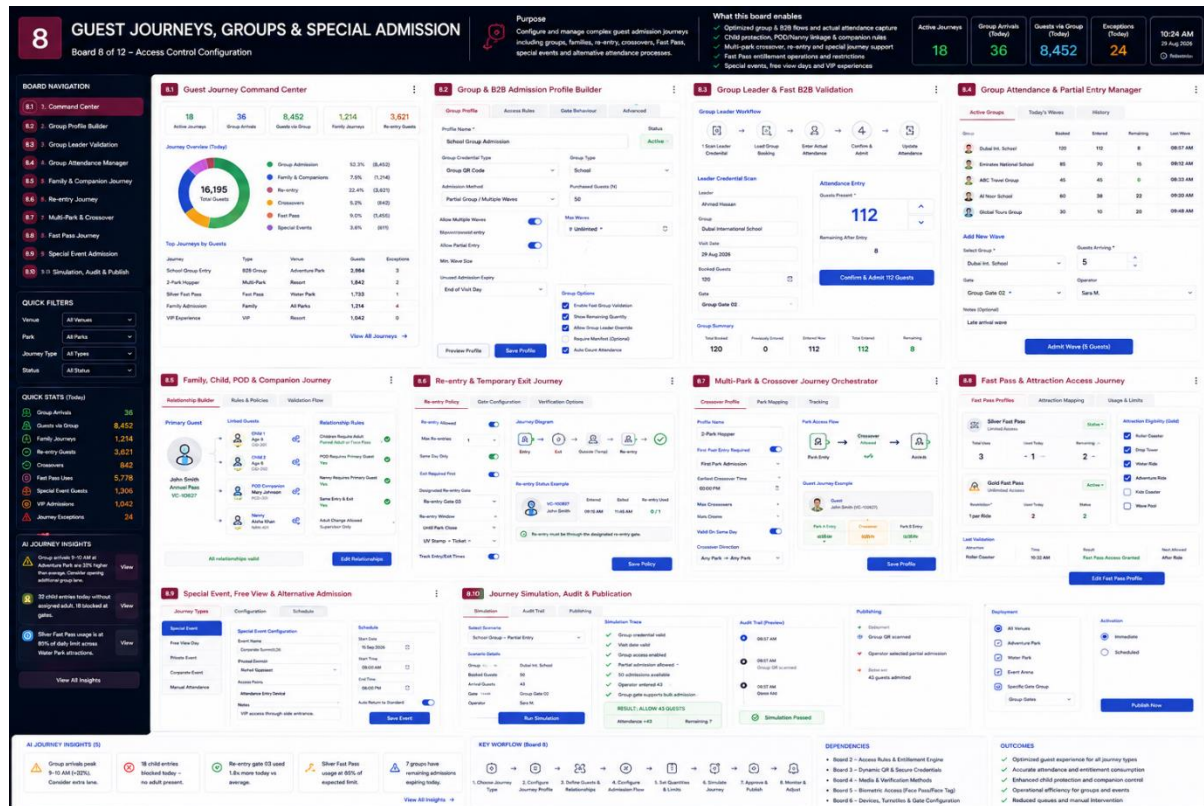
84 GATES OPERATING

with live local transactions accumulating and then a visual **reconnection/synchronization flow back to the cloud**. That will make the resilience architecture immediately understandable to both technical teams and venue executives.

Access Control Module — Board 8

Guest Journeys, Groups & Special Admission

Board 8 of 12 — 10 Backend Configuration Screens



Board 8 manages the **complex operational admission journeys** that cannot be handled as a simple "scan one ticket → admit one guest" transaction.

It covers the requirements for **B2B/group admission, family and child journeys, POD/nanny linkage, group QR codes, partial group entry, re-entry, multi-park crossover, Fast Pass, special-event attendance, free-view days, VIP/special access, and parking-linked arrival journeys**. The matrix specifically requires group tickets with N admissions, actual-attende capture, faster B2B group validation, adult-child pairing, POD/nanny binding, special-event attendance without standard physical admission, and different access processes on special dates.

Important separation from Board 2

Board 2 — Access Rules & Entitlement Engine

defines the underlying rules:

"This ticket allows 3 entries."

Board 8 defines the operational journey:

"A school group of 50 arrives with one group leader QR; 43 guests enter now, 7 arrive later, attendance must reflect 43, and the remaining 7 must stay available."

So Board 8 is the **Guest Journey Orchestration Layer**.

Screen 8.1 — Guest Journey Command Center

Purpose

Central configuration and monitoring screen for all special and multi-person admission journeys.

KPI Cards

Show:

- Active Journey Profiles
- Group Arrivals Today
- Guests via Group Admission
- Family Journeys
- Re-entry Guests
- Crossovers Today
- Fast Pass Validations
- Special Event Admissions
- VIP Admissions
- Journey Exceptions

Journey Portfolio

Journey	Type	Venue	Credential	Status
School Group Entry	B2B Group	Adventure Park	Group QR	Active
Family Admission	Family	All Parks	Mixed	Active
2-Park Hopper	Multi-Park	Resort	QR/RFID	Active
Silver Fast Pass	Fast Pass	Attractions	RFID	Active
VIP Experience	VIP	Resort	Face/QR	Active

Live Journey Health

Highlight:

- delayed groups
- unusually high manual intervention
- incomplete group entry
- companion violations
- crossover exceptions
- Fast Pass anomalies.

AI Journey Assistant

Example:

AI Alert: School groups arriving between 09:00–10:00 are generating an average 11-minute queue at Group Gate 2. Recommend opening an additional group-validation lane.

Acceptance Condition

Operations can centrally configure and monitor all non-standard guest admission journeys.

Screen 8.2 — Group & B2B Admission Profile Builder

Purpose

Configure admission for:

- Schools
- Tour Operators
- Corporate Groups
- Resellers
- Travel Groups
- Camps
- Families
- Events.

The matrix requires group products to support N admissions and one ticket scan capable of unlocking access for the configured number of people.

Group Profile

Example:

School Group Admission

Group Credential

Choose:

- Single Group QR
- Group Barcode
- Group RFID
- Group Leader Credential
- Individual Credentials
- Hybrid.

Admission Quantity

Purchased Guests

50

Admission Method

- Entire Group
- Partial Group
- Multiple Waves
- Individual Scan
- Leader + Quantity
- Manifest-Based.

Gate Behavior

One group scan can:

Authorize N Guests

Increment Attendance N

Open Group Gate

Acceptance Condition

Group admission behavior can be configured independently from standard individual-ticket admission.

Screen 8.3 — Group Leader & Fast B2B Validation

Purpose

Solve the specific matrix requirement for faster entrance flow when large B2B groups have multiple tickets stored on one device.

Group Leader

Configure:

Leader

Authorized

Credential

Group Leader QR

Group

Dubai International School

Booked

120 Guests

Fast Validation Journey

SCAN LEADER QR

↓

TICVAI retrieves booking/group

↓

GROUP: 120 GUESTS

Operator selects:

Admit All 120

or

Enter Actual Attendance

[112]

↓

CONFIRM

↓

Attendance:

+112

Remaining:

8

Bulk Credential Mode

Where individual tickets exist on the leader's device:

Load Group Booking

→ Show all associated tickets

→ Validate eligible tickets together

rather than requiring the operator to manually locate and present each ticket.

Pre-Arrival Status

Show:

- Payment ✓
- Booking ✓
- Visit Date ✓
- Group Product ✓
- Access Rules ✓
- Manifest ✓

Result:

READY FOR FAST ENTRY

Acceptance Condition

Large B2B groups can be admitted using a streamlined workflow without individually processing every credential when the configured group product allows it.

Screen 8.4 — Group Attendance & Partial Entry Manager

Purpose

Manage actual attendance when fewer guests arrive than the quantity purchased.

The matrix explicitly requires the scanner to show the exact group size and allow the operator to enter actual attendants so daily attendance is updated correctly.

Example

Group Booking

Purchased: 50

Previously Entered: 0

Guests Present:

[43]

Operator selects:

ADMIT 43

System records:

Purchased: **50**

Entered: **43**

Remaining: **7**

Later Arrival

Second scan:

Remaining:

7

Guests arriving:

5

Result:**Total Entered:**

48

Remaining:

2

Policy**Configure:****Allow Partial Admission**

YES

Allow Multiple Waves

YES

Maximum Waves

Unlimited / N

Unused Admission Expiry

End of Visit Day

Audit**Each wave records:**

- Group
- Leader
- Gate
- Operator
- Quantity
- Time
- Device.

Acceptance Condition

Attendance reflects the number of guests actually admitted rather than simply the quantity on the group ticket.

Screen 8.5 — Family, Child, POD & Companion Journey

Purpose

Configure linked-person access journeys.

The matrix requires child protection through adult-ticket pairing or biometric validation of the assigned adult. It also requires POD accompanying persons and nannies to be bound to a primary guest and only enter when accompanied by that guest.

Relationship Builder**PRIMARY GUEST**

John Smith

↓

Linked:**Child 1****Child 2****POD Companion****Nanny****Relationship Types****Configure:**

- Parent → Child
- Guardian → Minor
- POD → Companion
- Primary Guest → Nanny
- Group Leader → Group Member
- other authorized relationships.

Child Entry**Rule:**

Child requires Adult

Validation options:**Paired Adult Credential**

or

Assigned Adult Face Pass**Child Exit****Configure separately:**

Assigned Adult required for exit

This is important for child-protection journeys.

POD/Nanny**Example:****Nanny credential scanned alone:****YELLOW / DENY****Reason:****PRIMARY GUEST REQUIRED****Primary guest present:****ALLOW****Acceptance Condition**

Dependent and companion credentials cannot bypass their configured relationship requirements.

Screen 8.6 — Re-entry & Temporary Exit Journey**Purpose**

Manage guests temporarily leaving and returning to the venue.

The source matrix requires configurable re-entry and also describes a journey using a designated re-entry gate with both ticket verification and a UV stamp.

Re-entry Profile

Configure:

Re-entry Allowed

YES

Maximum

1

Same Day

YES

Exit Required First

YES

Designated Gate

Re-entry Gate 03

Temporary Exit

Guest scans:

EXIT

System status:

TEMPORARILY OUTSIDE

Optional Additional Verification

Configure:

- Credential only
- Credential + UV stamp
- Credential + Face
- Credential + operator
- custom supported method.

Re-entry

System checks:

- ✓ Previous entry
- ✓ Valid exit
- ✓ Re-entry entitlement
- ✓ Re-entry quantity
- ✓ Correct gate
- ✓ Anti-passback
- ✓ Additional verification

→

ALLOW RE-ENTRY

Acceptance Condition

Temporary exit and re-entry are tracked as distinct journey events rather than being counted as new normal admissions.

Screen 8.7 — Multi-Park & Crossover Journey Orchestrator

Purpose

Operationalize the multi-park rules configured in Board 2.

The matrix specifically distinguishes **crossover** from normal entry and re-entry and requires crossover to be tracked separately.

Visual Journey

PARK A

First Entry

↓

Guest leaves/transfers

↓

CROSSOVER

↓

PARK B

Configuration

Example:

2-Park Hopper

First Park:

Any

Second Park:

Water Park

Crossover:

Allowed after first park admission

Earliest crossover:

14:00

Maximum:

1

Journey Status

Credential:

VC-882173

Current:

Adventure Park — INSIDE

Eligible Next:

Water Park — CROSSOVER AVAILABLE

Tracking

Keep separate:

- Normal Entry
- Re-entry
- Crossover

- Exit.

AI

AI can detect:

Guest is attempting crossover to Park B without satisfying the configured first-entry requirement at Park A.

Acceptance Condition

Multi-park guest movement is orchestrated and reported independently from standard entry/re-entry.

Screen 8.8 — Fast Pass & Attraction Access Journey

Purpose

Configure the operational experience for limited and unlimited priority-access entitlements.

The matrix requires Silver Fast Pass to support three accesses and Gold to support unlimited access with an optional one-access-per-ride restriction.

Fast Pass Profiles

SILVER

Total Uses:

3

Consumption:

1 per validation

Remaining shown at scanner:

2 FAST PASSES REMAINING

GOLD

Total:

Unlimited

Restriction:

Maximum 1 per Ride

Attraction Mapping

Select eligible:

- Roller Coaster
- Drop Tower
- Water Ride
- Adventure Ride.

Validation

Guest scans at attraction:

Credential

↓

Fast Pass Entitlement

↓

Attraction Eligibility

↓

Usage Restriction

↓

Consume / Record

↓

FAST PASS ACCESS GRANTED**Operator Display****Show:****GOLD****Ride: Coaster****Previous Use: 10:32****Eligible: YES****Acceptance Condition**

Priority-access journeys correctly consume and display limited/unlimited entitlement usage.

Screen 8.9 — Special Event, Free View & Alternative Admission**Purpose**

Configure temporary/special admission processes that differ from normal venue access.

The matrix requires special-event products capable of capturing attendance without physical admission, *N*-person attendance entered through a turnstile/tablet/handheld, and Free View days where main gates are open while attraction gates continue validating tickets.

Journey Types**Configure:****Special Event****Free View Day****Private Event****Corporate Event****School Event****Manual Attendance****Open Main Gate / Paid Attractions****Example — Free View Day****Main Entrance****Validation:**

OFF

Mode:

FREE SPIN

Attendance:

COUNT PHYSICAL PASSAGE

Attractions**Validation:**

ON

Entitlement:

REQUIRED

Example — Special Event

Product:

Corporate Event Attendance

Operator enters:

85 Guests

Attendance:

+85

Physical admission:

Not Required

Audit Classification:

SPECIAL EVENT ADMISSION

Scheduled Activation

Configure:

15 Sep 2026

08:00–18:00

At end:

Automatically return to Standard Access Configuration

Acceptance Condition

Special admission processes can temporarily replace normal access behavior without permanent gate reconfiguration.

Screen 8.10 — Journey Simulation, Audit & Publication

Purpose

Test an entire guest journey—not merely an individual scan—before deploying it.

Simulation Scenarios

Support:

- Group of 50 / 43 attending
- Group arriving in three waves
- Group leader validation
- Child without adult
- Child + assigned adult
- POD companion without primary guest
- Temporary exit + re-entry
- Incorrect re-entry gate
- Multi-park crossover
- Silver Fast Pass exhausted

- Gold Fast Pass repeated ride
- Free View Day
- Special Event attendance.

Example

Scenario

School Group

Purchased:

50

Arrive:

43

Credential:

Group QR

Gate:

Group Gate 02

Simulation Trace

- ✓ Group credential valid
- ✓ Visit date valid
- ✓ Group access enabled
- ✓ Partial admission allowed
- ✓ 50 admissions available
- ✓ Operator entered 43
- ✓ Group gate supports bulk admission

RESULT

ALLOW 43 GUESTS

Attendance:

+43

Remaining:

7

Journey Audit

Audit should show the entire sequence:

08:57 Group QR scanned

08:57 Operator selected partial admission

08:58 43 guests admitted

09:31 Group QR rescanned

09:32 5 additional guests admitted

rather than only independent scan records.

AI Journey Conflict Check

Detect:

- impossible journey sequences
- missing gates
- incompatible hardware

- missing companion relationship
- conflicting quantities
- duplicate attendance
- journey/rule mismatch.

Publication

Draft → Simulate → Validate → Approval → Schedule → Publish

Deployment scope:

- Tenant
- Venue
- Park
- Journey
- Product
- Group type
- Event.

Acceptance Condition

Complex guest journeys can be simulated end-to-end, approved, deployed, audited and rolled back.

Board 8 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
8.1	Guest Journey Command Center	Special journey management and monitoring
8.2	Group & B2B Admission Profile Builder	Configure group-access models
8.3	Group Leader & Fast B2B Validation	Accelerate large-group entry
8.4	Group Attendance & Partial Entry Manager	Actual attendance and multiple arrival waves
8.5	Family, Child, POD & Companion Journey	Linked-person and child-protection journeys
8.6	Re-entry & Temporary Exit Journey	Temporary exit and controlled re-entry
8.7	Multi-Park & Crossover Journey Orchestrator	Cross-park movement
8.8	Fast Pass & Attraction Access Journey	Priority-access entitlement operations
8.9	Special Event, Free View & Alternative Admission	Exceptional admission processes
8.10	Journey Simulation, Audit & Publication	End-to-end testing and governance

Board 8 key workflow

Select Journey Type → Select Product/Credential → Define Guest Relationship/Group → Configure Admission Method → Define Quantities → Configure Gate Journey → Configure Exceptions → Simulate Complete Journey → Approve → Publish

One thing I would deliberately NOT put into Board 8

Parking configuration itself should not become an Access Control submodule. The matrix requires integration with parking-management software and prepayment during ticket purchase. Access Control should therefore receive and validate the resulting **parking entitlement/credential**, while parking pricing, inventory, payment and parking-space operations remain with their respective modules/integration.

This keeps Board 8 focused on **people and guest admission journeys** and prevents it from turning into another general operational module.

Access Control Module — Board 9

Live Access Operations, Podium & Exception Management

Board 9 of 12 — 10 Backend Configuration Screens



Board 9 is the **real-time operational control layer** of TICVAI Access Control.

Boards 1–8 configure the access environment, rules, credentials, hardware, offline operation, and guest journeys. Board 9 is where the **venue operations team actually controls admission during live operations**.

It covers the matrix requirements around:

- Podium operations
- Operator login and accountability
- Ticket lookup
- Manual validation and overrides
- Gate/turnstile mode changes
- Invalid reason codes
- Manual ticket disabling
- Blacklist/whitelist management
- Operational exceptions
- Gate intervention
- Live queue and throughput monitoring
- Emergency operational actions
- Complete operator audit trail

The core principle is:

Automation should process normal admissions. Board 9 manages the exceptions.

The normal guest should pass without operator involvement. The operational console becomes important when TICVAI returns **Yellow**, a guest has a problem, a group needs assistance, a supervisor must override a rule, or venue operations need to change gate behavior.

Screen 9.1 — Live Access Operations Command Center

Purpose

Provide the venue control room with a **real-time view of access operations across all gates, parks, zones, and attractions.**

Global KPI Cards

Display:

- Guests Entered Today
- Guests Exited
- Guests Currently In Park
- Valid Scans
- Rejected Scans
- Yellow / Intervention Scans
- Overrides
- Active Gates
- Offline Gates
- Average Validation Time
- Guests / Minute
- Active Operational Alerts

Live Venue Map

Use the access topology from Board 1.

Example:

Adventure Park

-  Main Gate 01
-  Main Gate 02
-  Main Gate 03
-  Main Gate 04
-  VIP Gate
-  Group Gate
-  Re-entry Gate

Selecting a gate opens its operational status.

Gate Card

MAIN GATE 03

Mode: ENTRY

Status: ONLINE

Queue: Moderate

Throughput: 31 Guests/min

Last Scan: 4 sec ago

Valid: 92%

Yellow: 5%

Rejected: 3%

Operational Alerts

Examples:

QUEUE WARNING

Main Entrance wait time exceeds configured target.

REJECTION SPIKE

Gate 07 rejection rate increased from 2.1% to 14.8%.

DEVICE WARNING

VIP Gate 02 has entered offline mode.

AI Operations Assistant

AI can recommend:

Open Gates 09–12. Current Main Entrance demand is projected to exceed available throughput within 8 minutes.

or:

61% of current rejection events relate to incorrect visit dates. Consider assigning one guest-service operator before the scan line.

Acceptance Condition

Operations can understand the live access state of the entire venue without opening individual device screens.

Screen 9.2 — Podium Operations Console

Purpose

Provide the operational interface described in the matrix for attendants controlling one or more turnstiles.

The podium may be:

- physical keyboard/control panel
- tablet
- workstation
- handheld
- other authorized operational interface.

Podium Assignment

Example:

Podium — Main Entrance A

Controls:

Gates 01–06

Operator

Sara M.

Role:

Access Supervisor

Shift:

08:00–16:00

Live Gate Tiles

Display:

Gate 01 — ENTRY 

Gate 02 — ENTRY 

Gate 03 — ENTRY 

Gate 04 — ENTRY 

Gate 05 — CLOSED 

Gate 06 — GROUP 

Quick Actions

Authorized actions:

- Ticket Lookup
- Rescan
- Manual Validate
- Override
- Open Gate
- Change Gate Mode
- Group Admission
- View Ticket History
- Disable Credential
- Escalate Issue.

Important

The podium does **not define access rules**.

It executes operational actions against rules configured elsewhere.

Acceptance Condition

Authorized attendants can control assigned access devices from one operational interface.

Screen 9.3 — Ticket & Credential Investigation Console

Purpose

Allow operators to quickly investigate why a guest cannot enter.

Search Methods

Search by:

- Ticket ID
- QR
- Barcode
- RFID

- NFC
- Virtual Credential ID
- Order Number
- Membership
- Guest Name
- Email
- Mobile
- External Partner Reference.

Credential Summary

Example:

VC-108427

Adventure Park Day Ticket

Guest: John Smith

Visit Date: 01 Sep 2026

Status:

ACTIVE

Current Journey

Entry:

09:42

Exit:

—

Current Location:

IN PARK

Re-entry:

1 Remaining

Entitlements

Park Entry: USED

Fast Pass: 2 / 3 Remaining

Meal Voucher: AVAILABLE

Locker: L-284

Verification Method

Dynamic QR — LOCKED

Access History

Time	Gate	Action	Result
09:42	Main 04	Entry	Allowed
10:31	Ride 07	Fast Pass	Allowed
11:18	Ride 03	Fast Pass	Allowed

Related Transaction

Where permitted, show ticket transaction context:

- Transaction time
- Sales channel
- POS
- Clerk
- Payment reference/method
- Reissue history.

Acceptance Condition

An operator can understand the ticket's complete operational access state from one screen without navigating across multiple modules.

Screen 9.4 — Validation Exception & Reason Code Manager

Purpose

Standardize what happens when access is not automatically granted.

The matrix requires the scanner to display a reason code when a ticket is invalid.

Reason Code Library

Examples:

AC-001 — Ticket Expired

AC-002 — Wrong Visit Date

AC-003 — Already Used

AC-004 — Wrong Park

AC-005 — Re-entry Not Allowed

AC-006 — Anti-Passback

AC-007 — Credential Revoked

AC-008 — Verification Method Mismatch

AC-009 — Adult Companion Required

AC-010 — Biometric Review Required

AC-011 — Fast Pass Exhausted

AC-012 — Ticket Blacklisted

Configuration

For each reason:

Guest Message

This ticket is not valid for today's visit.

Operator Message

Ticket date: 02 Sep 2026

Outcome

Choose:

☒ Deny

- ☐ Operator Review
- ☐ Supervisor Required
- ☒ Allow with Warning

Suggested Resolution

Example:

Wrong Date →

Check Reschedule Eligibility

Already Used →

View Previous Scan

Verification Mismatch →

Verify Guest Identity

Acceptance Condition

Every validation failure returns a standardized reason and configured operational response.

Screen 9.5 — Manual Override & Supervisor Approval

Purpose

Allow authorized staff to bypass selected access restrictions when operationally justified.

The matrix explicitly requires **Allow Override** and operator-based ticket override.

Override Journey

Guest scans.

Result:

ACCESS DENIED

Reason:

WRONG VISIT DATE

Operator selects:

REQUEST OVERRIDE

Override Panel

Show:

Ticket:

VC-108427

Configured Date:

02 Sep

Current Date:

01 Sep

Gate:

Main Gate 03

Operator Reason

Select:

- Guest Service
- Ticketing Error
- Operational Exception
- Management Authorization
- Technical Failure
- Event Exception
- Other.

Approval Policy

Example:

Wrong Date:

Supervisor Approval Required

Anti-Passback:

Supervisor Approval

Expired Ticket:

Not Overrideable

Fraud Lock:

Security Approval Required

Approval

Supervisor:

Ahmed K.

Select:

APPROVE ONE-TIME ENTRY

Result

OVERRIDE GRANTED

Gate opens.

Critical Requirement

The original validation decision must remain unchanged in the audit trail.

Store:

Original Result: DENIED

Override: APPROVED

This avoids rewriting history as though the ticket was originally valid.

Acceptance Condition

Only authorized users can override eligible rules, with reason, approval, timestamp, operator, device and original decision retained.

Screen 9.6 — Credential Disable, Blacklist & Whitelist Operations

Purpose

Provide immediate operational security control over individual credentials.

The matrix requires:

- blacklist/whitelist capability
- manual ticket invalidation
- manual disabling of individual guest access.

Search Credential

Example:

VC-983427

Status:

ACTIVE

Actions

DISABLE ACCESS

Reason:

- Lost Ticket
- Stolen Credential
- Fraud Suspected
- Guest Removal
- Security Incident
- Duplicate Credential
- Management Instruction
- Other.

Scope

Disable:

- Entire Credential
- Venue Access
- Attraction Access
- Re-entry
- Fast Pass
- Specific Entitlement.

Duration

- Permanent
- Until End of Day
- Until Date/Time
- Until Manually Restored.

Blacklist

Add:

Credential

to:

ACCESS BLACKLIST

Distribution:

- ✓ Central Platform
- ✓ Venue Edge
- ✓ Online Gates
- ✓ Offline Revocation Package

Whitelist

Where business/security rules permit, authorized security administrators may explicitly whitelist approved exceptions.

High-Risk Action

Blacklisting/whitelisting should support:

Supervisor/Security Approval

depending on policy.

Acceptance Condition

Authorized users can immediately restrict credential access across the configured venue environment.

Screen 9.7 — Live Gate Mode & Lane Control

Purpose

Allow operations to change access-point modes during live operations without entering the Board 6 engineering configuration.

Board 6 defines which modes a device **can support**.

Board 9 controls which permitted mode it is **currently running**.

Gate Grid

Select:

- ☒ Gate 01
- ☒ Gate 02
- ☒ Gate 03
- ☒ Gate 04

Current:

ENTRY

Change Mode

Available:

- Entry
- Exit
- Re-entry
- Crossover
- Group
- Fast Pass
- Free Spin
- Count Only
- Closed.

Example

At 17:30:

Select Gates 08–12.

Change:

ENTRY

→

EXIT

Bulk Command

5 GATES SELECTED

APPLY EXIT MODE

Confirmation

Show:

- affected gates
- current mode
- target mode
- operator
- reason
- effective time.

Scheduled Mode

Optional:

Switch Gates 08–12 to EXIT at 17:30

Emergency

Emergency/Drop Arm actions remain available to specifically authorized users, consistent with Board 6 safety controls.

Acceptance Condition

Operations can change the active mode of one or multiple authorized gates without modifying their underlying hardware configuration.

Screen 9.8 — Queue, Throughput & Lane Optimization

Purpose

Manage entrance flow in real time.

This addresses the matrix requirement to improve entrance flow and queuing, particularly for large B2B groups.

Live Entrance View

MAIN ENTRANCE

Guests Waiting:

486

Active Lanes:

12 / 16

Current Throughput:

318 guests / 10 min

Estimated Wait:

14 min

Lane Performance

Lane	Type	Guests/min	Reject %	Status
G01	Standard	28	2.1%	Healthy
G02	Standard	31	1.8%	Healthy
G03	Group	47	0.9%	Healthy
G04	Standard	12	14.2%	Investigate

Journey Queues**Show:**

- Standard
- Family
- Group/B2B
- VIP
- POD/Accessible
- Re-entry
- Fast Pass.

AI Lane Optimization**AI recommendation:**

Convert Gates 10 and 11 from Standard Entry to Group Entry for the next 25 minutes.

Reason:

4 scheduled B2B groups / 327 guests arriving.

Operator Action**APPLY RECOMMENDATION**

This must still respect operator permissions and hardware capabilities.

Forecast**Show projected:**

15 min

30 min

60 min

using scheduled arrivals and current flow.

Acceptance Condition

Operations can monitor throughput and proactively adjust available lane capacity.

Screen 9.9 — Operational Incident & Exception Workspace

Purpose

Manage access incidents that require more than a simple override.

Incident Types

- Credential Fraud
- Duplicate Use
- Biometric Mismatch
- Lost Wristband
- Gate Failure
- Guest Dispute
- Child/Guardian Issue
- Group Admission Issue
- Security Event
- Offline Conflict
- Partner Ticket Failure
- Emergency Access Event.

Create Incident

Example:

Incident AC-2026-00842

Guest:

John Smith

Credential:

VC-108427

Gate:

Main Gate 03

Issue:

Repeated credential use

Evidence

Automatically attach:

- Scan history
- reason codes
- gate/device
- operator
- ticket status
- credential history
- access journey
- relevant security alerts.

Assignment

Assign to:

- Access Supervisor
- Guest Services
- Security
- Ticketing
- Technical Support.

Status

Open → Investigating → Resolved → Closed

Resolution

Example:

Credential duplicated

Action:

Original credential retained

Duplicate media revoked

AI Investigation Assistant

AI can summarize:

Credential was successfully used at Gate 04 at 09:42. A second device attempted entry at Gate 09 at 09:44 and Gate 11 at 09:45. Device-binding mismatch was detected on both attempts.

Acceptance Condition

Complex access exceptions can be formally investigated and resolved with complete operational evidence.

Screen 9.10 — Operations Audit, Shift Handover & Control Summary

Purpose

Provide full accountability for everything operators and supervisors changed during live access operations.

Operator Audit

Record:

- Operator
- Role
- Podium
- Device
- Login
- Logout
- Ticket lookups
- Overrides
- Manual openings
- Mode changes
- Credential disables
- Blacklist changes
- Group adjustments
- incidents.

Shift Summary

Example:

Morning Shift — Main Entrance

08:00–16:00

Guests Entered: 18,426

Rejected: 382

Yellow Interventions: 641

Overrides: 43

Manual Gate Opens: 17

Incidents: 8

Offline Events: 2

Handover

Outgoing supervisor records:

Open Issues

- Gate 08 RFID intermittent
- School group expected 16:15
- Credential fraud incident under investigation
- Re-entry Gate 02 temporarily closed.

Incoming supervisor:

ACCEPT HANDOVER

Exception Summary

Automatically highlight:

- unusual override volumes
- unresolved incidents
- disabled gates
- blacklisted credentials
- offline devices
- abnormal rejection rates.

AI Shift Summary

AI generates a concise operational summary:

Main entrance operated normally overall. Gate 08 generated 38% of RFID read failures. Override activity was concentrated between 10:10–10:45 due to a ticket-date configuration issue affecting one reseller batch.

Export / Reporting

Provide:

- Shift Report
- Operator Activity Report
- Override Report
- Gate Mode Change Report
- Incident Report

- Exception Report.

Acceptance Condition

Every operational intervention is attributable to a user and can be reviewed during audit, investigation and shift handover.

Board 9 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
9.1	Live Access Operations Command Center	Real-time venue admission operations
9.2	Podium Operations Console	Operator control of gates/turnstiles
9.3	Ticket & Credential Investigation Console	Investigate guest access state
9.4	Validation Exception & Reason Code Manager	Standardize failed/yellow validation outcomes
9.5	Manual Override & Supervisor Approval	Govern access overrides
9.6	Credential Disable, Blacklist & Whitelist Operations	Immediate credential security controls
9.7	Live Gate Mode & Lane Control	Change active gate operating modes
9.8	Queue, Throughput & Lane Optimization	Optimize guest flow
9.9	Operational Incident & Exception Workspace	Investigate complex access problems
9.10	Operations Audit, Shift Handover & Control Summary	Accountability and operational continuity

Board 9 Key Workflow

Monitor Live Operations → Detect Exception → Investigate Credential → Understand Reason → Resolve / Override / Escalate → Adjust Gate or Lane if Required → Record Incident → Continue Guest Journey → Audit & Handover

Important design distinction

For **Board 9 design**, I would make it visually very different from the configuration-heavy Boards 1–8.

This board should feel like a genuine **Access Control Operations Control Room**:

LIVE VENUE MAP + GATE STATUS + GUEST FLOW + ALERTS + OPERATOR ACTIONS + QUEUES + EXCEPTIONS

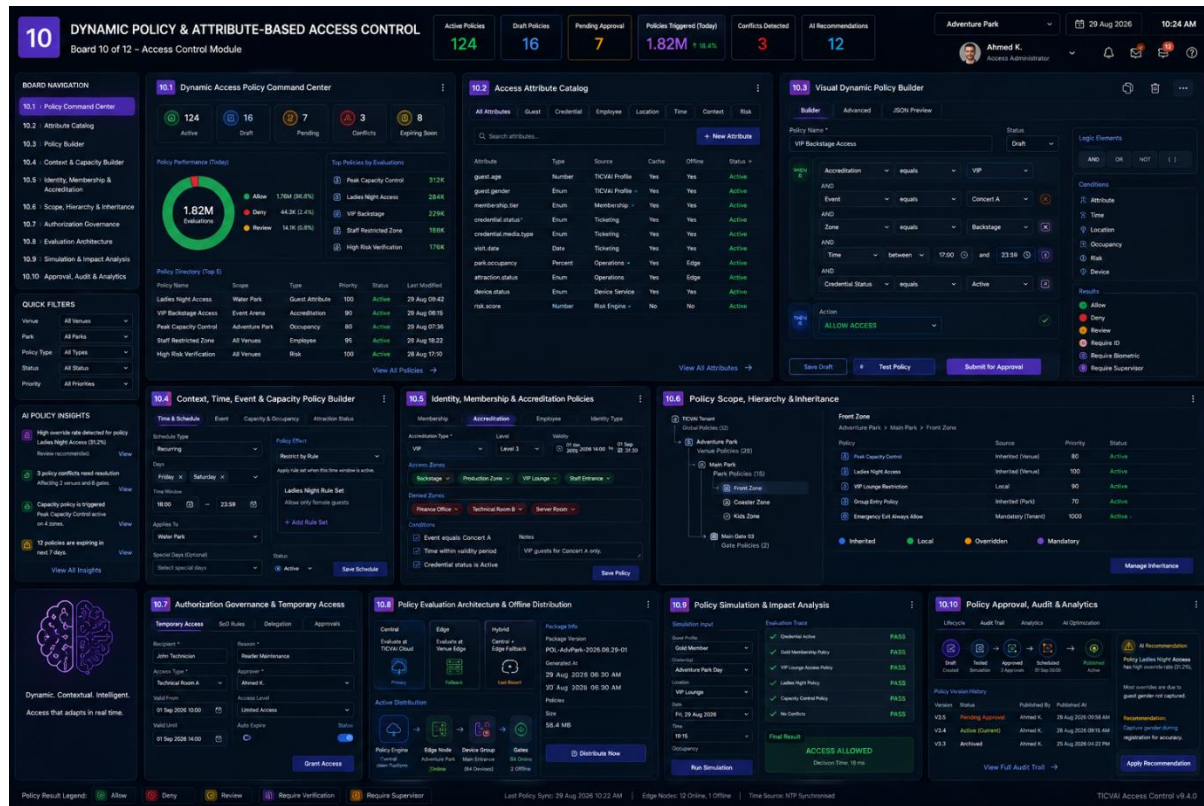
The operator should immediately see **Green / Yellow / Red** conditions across the venue, while sensitive actions such as **Override, Blacklist, Manual Open, and Gate Mode Change** are visually controlled and permission-aware.

That distinction is important: **Boards 1–8 configure how TICVAI should behave; Board 9 operates TICVAI while guests are actually entering the venue.**

Access Control Module — Board 10

Dynamic Policy & Attribute-Based Access Control

Board 10 of 12 — 10 Backend Configuration Screens



Board 10 is the **advanced policy intelligence layer** of TICVAI Access Control.

While **Board 2 — Access Rules & Entitlement Engine** answers questions such as:

“Does this ticket have the entitlement required to enter this attraction?”

Board 10 handles more contextual decisions:

“Considering who this person is, where they are, when they are requesting access, their credential status, membership/accreditation, venue conditions, capacity, risk level, and other attributes — should access be granted right now?”

This is essentially TICVAI's **Attribute-Based Access Control / Dynamic Policy Engine (ABAC)**.

Core Decision Model

WHO

User / Guest / Employee / Member / Accreditation

•

WHAT

Credential / Ticket / Membership / Entitlement

•

WHERE

Venue / Park / Zone / Attraction / Gate / Resource

•

WHEN

Date / Time / Event / Season / Operating Period

•

CONTEXT

Capacity / Occupancy / Risk / Device / Journey / Status

↓

TICVAI POLICY ENGINE

↓

ALLOW / DENY / REVIEW / REQUIRE ADDITIONAL VERIFICATION**Screen 10.1 — Dynamic Access Policy Command Center****Purpose**

Provide central governance and visibility over all dynamic access policies.

KPI Cards**Show:**

- Active Policies
- Draft Policies
- Policies Pending Approval
- Policies Triggered Today
- Allow Decisions
- Deny Decisions
- Review Decisions
- Policy Conflicts
- Expiring Policies
- **AI Recommendations**

Policy Directory

Policy	Scope	Type	Priority	Status
Ladies Night Access	Water Park	Guest Attribute	100	Active
VIP Backstage	Event Arena	Accreditation	90	Active
Peak Capacity Control	Adventure Park	Occupancy	80	Active
Staff Restricted Zone	Resort	Employee	95	Active
High Risk Verification	All Venues	Risk	100	Active

Policy Health**Display:****Healthy****Conflict****Unused**

High Denial Rate**Expiring****Requires Review****AI Policy Insights****Example:**

Policy Conflict: VIP Backstage Access allows VIP accreditation while Restricted Production Zone Policy denies the same user group between 18:00–19:00.

Acceptance Condition

Administrators can centrally understand the status, scope, performance and conflicts of all dynamic access policies.

Screen 10.2 — Access Attribute Catalog

Purpose

Define the attributes available to the TICVAI policy engine.

This becomes the reusable **data dictionary** for dynamic access decisions.

Attribute Categories**Guest Attributes**

- Age
- Gender where legally/business permitted
- Customer Segment
- Guest Category
- Country/Residency where applicable
- Group Type
- Companion Relationship
- POD status

Credential Attributes

- Ticket Type
- Credential Status
- Media Type
- Verification Method
- Visit Date
- Entitlements
- Membership
- Accreditation
- Loyalty Tier

Employee Attributes

- Employee
- Department
- Role

- Job Function
- Accreditation
- Shift
- Security Clearance

Location Attributes

- Venue
- Park
- Zone
- Attraction
- Gate
- Resource
- Restricted Area

Time Attributes

- Date
- Time
- Day
- Season
- Event
- Performance
- Operating Calendar
- Special Day

Operational Attributes

- Capacity
- Occupancy
- Queue Level
- Attraction Status
- Venue Status
- Emergency State

Device Attributes

- Device Type
- Device ID
- Device Security Status
- Online/Offline
- Verification Capability
- Location

Risk Attributes

- Credential Risk
- Device Risk
- Guest Risk

- Fraud Score
- Behavioral Anomaly.

Attribute Definition

Example:

Attribute

membership.tier

Type:

ENUM

Values:

Standard / Silver / Gold / Platinum

Source:

TICVAI Membership

Cache:

Allowed

Offline:

Yes

Acceptance Condition

Policy designers use governed attributes rather than manually creating inconsistent fields in individual policies.

Screen 10.3 — Visual Dynamic Policy Builder

Purpose

Provide a no-code interface for constructing contextual access policies.

Policy Example

VIP BACKSTAGE ACCESS

IF

Accreditation = VIP

AND

Event = Concert A

AND

Zone = Backstage

AND

Time BETWEEN 17:00 AND 23:59

AND

Credential Status = Active

THEN

ALLOW ACCESS

ELSE

DENY

Visual Builder**Use blocks:****WHEN****AND****OR****NOT****IN****BETWEEN****GREATER THAN****LESS THAN****MATCH****Actions****Policy result can be:**☒ **ALLOW**☒ **DENY**☐ **REVIEW**☐ **REQUIRE ID**☐ **REQUIRE BIOMETRIC**☐ **REQUIRE COMPANION**☐ **REQUIRE SUPERVISOR****Nested Logic****Example:**

Membership = Gold

AND

(
Occupancy < 90%

OR

VIP Override = True

)

Acceptance Condition

Business administrators can create sophisticated contextual access policies without development work.

Screen 10.4 — Context, Time, Event & Capacity Policy Builder**Purpose**

Configure policies driven by changing venue conditions rather than only guest attributes.

Temporal Conditions**Support:**

- Date

- Day
- Time
- Season
- Event
- Performance
- Special Event
- Holiday
- Operating Calendar.

Example**Ladies Night**

Friday

18:00–23:59

↓

Apply:

Ladies Night Access Policy

Capacity Conditions**Example:****IF**

Zone Occupancy $\geq$ 90%

THEN

Stop Standard Admission

BUT

Allow Exit

AND

Allow Emergency/Security Roles

Attraction Conditions**If:**

Attraction Status = Closed

Then:

Deny Attraction Entry

Event Conditions**If:**

Event = Private Corporate Event

Then:**Only:**

Corporate Accreditation

or

Event Credential

may enter specified zones.

Dynamic Capacity Thresholds

Example:

0–79%

NORMAL

80–89%

MONITOR

90–94%

RESTRICT

95%+

STOP ADMISSION

Acceptance Condition

Access behavior can dynamically respond to operating calendars, events, capacity, occupancy and venue status.

Screen 10.5 — Identity, Membership & Accreditation Policies

Purpose

Configure policies based on who the requesting person is.

Identity Types

Support:

- Guest
- Member
- Annual Pass Holder
- Employee
- Contractor
- Vendor
- Performer
- Media
- VIP
- Security
- Emergency Services
- Event Staff.

Membership Policy

Example:

Gold Member

Can access:

- ✓ Priority Entrance
- ✓ Member Lounge
- ✓ Selected Attractions

During:

Membership Validity

Accreditation Policy

Example:

EVENT CREW — LEVEL 3

Allowed:

- ✓ Production Zone
- ✓ Backstage
- ✓ Staff Entrance

Denied:

- ✗ VIP Hospitality
- ✗ Finance Office

Employee Policy

Conditions:

Employee Status = Active

AND

Current Shift = Active

AND

Role allows zone

Temporary Accreditation

Valid:

01 Sep 14:00

until:

01 Sep 23:30

Automatically expires afterward.

Acceptance Condition

Memberships, staff identities and accreditations can participate directly in access-policy decisions.

Screen 10.6 — Policy Scope, Hierarchy & Inheritance

Purpose

Govern how policies apply across TICVAI's multi-tenant and multi-venue architecture.

Policy Hierarchy

Tenant

↓

Venue

↓

Park

↓

Zone

↓

Attraction

↓

Access Point

↓

Gate**Example****Tenant Policy:**

Credential must be Active

applies to:**ALL VENUES****Venue Policy:**

Water Park operating hours

applies only:**WATER PARK****Zone Policy:**

VIP Lounge restriction

applies:**VIP LOUNGE****Inheritance****Show:****Inherited****Local****Overridden****Mandatory / Cannot Override****Cross-Venue Policy****Example:****Global Fraud Lock****Scope:**

ALL TICVAI VENUES

This should override local access permissions.

Policy Priority**Example:**

1. Emergency Policy
2. Security/Fraud Policy
3. Regulatory/Safety Policy
4. Venue Restriction
5. Accreditation
6. Membership
7. Standard Access.

The actual hierarchy should be configurable rather than hard-coded.

Conflict Resolution

Administrator defines:

- Highest Priority Wins
- Deny Overrides Allow
- Most Specific Wins
- Mandatory Parent Wins
- Explicit Resolution.

Acceptance Condition

Policies behave predictably across tenants, venues and access hierarchies.

Screen 10.7 — Authorization Governance & Temporary Access

Purpose

Apply enterprise governance principles to privileged and temporary access.

Governance Controls

Support:

- Least Privilege
- Segregation of Duties
- Temporary Access
- Expiring Grants
- Delegated Administration
- Approval Requirements.

Temporary Grant

Example:

Contractor

Access:

Technical Room A

Date:

01 Sep

Time:

10:00–14:00

Reason:

Reader Maintenance

Approver:

Venue Security Manager

At:

14:00

ACCESS AUTOMATICALLY EXPIRES

Segregation of Duties**Example:****A user who:**

Creates Security Policy

cannot:

Approve Same Policy

Delegated Administration**Example:****Water Park Access Manager can modify:**

✓ Water Park operational policies

Cannot modify:

✗ Resort Global Security Policy

✗ Adventure Park Policies.

Emergency Temporary Access**Authorized users can issue:****Emergency Access Grant****with:**

- scope
- duration
- reason
- approver
- full audit.

Acceptance Condition

Privileged and temporary access follows controlled authorization rather than informal manual permissions.

Screen 10.8 — Policy Evaluation Architecture & Offline Distribution**Purpose**

Define where and how policies are evaluated.

This screen connects Board 10 with the offline/edge architecture already configured in Board 7.

Evaluation Modes**Policy can be:****CENTRAL**

Evaluated by TICVAI central policy service.

EDGE

Evaluated at venue edge.

DEVICE

Evaluated locally where supported.

HYBRID

Central when available, cached/edge fallback when offline.

Policy Compatibility

Policy	Central	Edge	Device	Offline
Ticket Status	✓	✓	✓	✓
Time Rule	✓	✓	✓	✓
Membership Tier	✓	✓	✓	✓
Live Occupancy	✓	✓	—	Conditional
Live Fraud AI	✓	—	—	No

Offline Fallback

Example:

Policy uses:

Live Risk Score

Central unavailable.

Fallback:

Cached Risk Score

Maximum age:

15 min

If unavailable:

REQUIRE OPERATOR REVIEW

Distribution

Central Policy Engine

↓

Venue Edge

↓

Authorized Device Groups

Acceptance Condition

Every policy has a defined evaluation location and predictable behavior when connectivity or data sources are unavailable.

Screen 10.9 — Policy Simulation, Conflict & Impact Analysis

Purpose

Test policies before they affect live guest admission.

Simulation Input

Example:

Guest:

Gold Member

Ticket:

Adventure Park

Location:

VIP Lounge

Date:

Friday

Time:

19:15

Occupancy:

82%

Risk:

Low

Evaluation Trace**Policy 1**

Credential Active

✓ PASS

Policy 2

Gold Membership

✓ PASS

Policy 3

VIP Lounge Access

✓ PASS

Policy 4

Capacity Restriction

✓ PASS

FINAL RESULT

 **ACCESS ALLOWED**

Conflict Detection**Example:****Policy A:**

Gold Members → Allow

Policy B:

Private Event → Deny Non-Event Credentials

Result:**POLICY CONFLICT****Show:**

- policies involved
- hierarchy
- priority

- resulting decision
- affected products
- affected venues
- estimated guests impacted.

Impact Analysis

Before publication:

This policy is expected to affect **12,480 active credentials** across **3 venues**.

Regression Testing

Run saved scenarios against new policy version.

Example:

148 scenarios

✓ 145 Passed

⚠ 3 Changed Outcome

Acceptance Condition

No policy reaches production without visibility into conflicts and potential operational impact.

Screen 10.10 — Policy Approval, Audit, Analytics & AI Optimization

Purpose

Govern the complete policy lifecycle and continuously measure policy effectiveness.

Lifecycle

Draft

→

Test

→

Impact Analysis

→

Approval

→

Schedule

→

Publish

→

Monitor

→

Optimize

→

Retire

Approval

Support:

- Policy Owner
- Business Approval
- Security Approval
- Technical Approval
- Dual Approval for sensitive policies.

Version History**Example:****VIP Access Policy**

V3.4 — Production

V3.5 — Pending Approval

V3.3 — Archived

Audit**Record:**

- Created by
- Modified by
- Approved by
- Changed fields
- Previous values
- New values
- Publication date
- affected scope.

Policy Analytics**Show:****Evaluations**

1.82M

Allow

96.8%

Deny

2.4%

Review

0.8%

Average Decision

18 ms

Effectiveness**Measure:**

- Denial Rate
- Override Rate
- False-Positive Indicators

- Operator Review Rate
- Guest Impact
- Security Incidents
- Policy Latency.

AI Policy Optimization

AI can identify:

Policy AC-184 generates a 31% operator override rate. Most overrides involve Gold Members entering VIP Lounge between 18:00–20:00. Review membership eligibility conditions.

AI may recommend:

MODIFY CONDITION

but **must not autonomously publish sensitive access-policy changes**.

Recommended model:

AI Recommend → Human Review → Approval → Publish

Rollback

Allow:

ROLLBACK TO V3.4

with full audit.

Acceptance Condition

Dynamic access policies are versioned, approved, measurable, auditable and continuously optimizable.

Board 10 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
10.1	Dynamic Access Policy Command Center	Policy estate and health
10.2	Access Attribute Catalog	Govern reusable decision attributes
10.3	Visual Dynamic Policy Builder	Build ABAC policies without code
10.4	Context, Time, Event & Capacity Policy Builder	Context-aware operational policies
10.5	Identity, Membership & Accreditation Policies	Identity-based authorization
10.6	Policy Scope, Hierarchy & Inheritance	Multi-venue policy governance
10.7	Authorization Governance & Temporary Access	Least privilege, SoD and temporary grants
10.8	Policy Evaluation Architecture & Offline Distribution	Central/edge/device evaluation
10.9	Policy Simulation, Conflict & Impact Analysis	Pre-production testing
10.10	Policy Approval, Audit, Analytics & AI Optimization	Lifecycle governance and intelligence

Board 10 Key Workflow

Define Attributes → Build Policy → Add Context → Select Identity/Accreditation Conditions → Define Scope & Inheritance → Apply Governance → Select Evaluation Architecture → Simulate & Detect Conflicts → Assess Impact → Approve → Publish → Monitor → AI Optimize

Critical distinction from Board 2

I would keep this separation very strict in the UI/UX:

Board 2 = Access Entitlement Engine

WHAT is this credential entitled to do?

Board 10 = Dynamic Policy Engine

UNDER THE CURRENT CONTEXT, should that entitlement be permitted right now?

For example:

Board 2:

Gold Member → VIP Lounge entitlement = YES

Board 10:

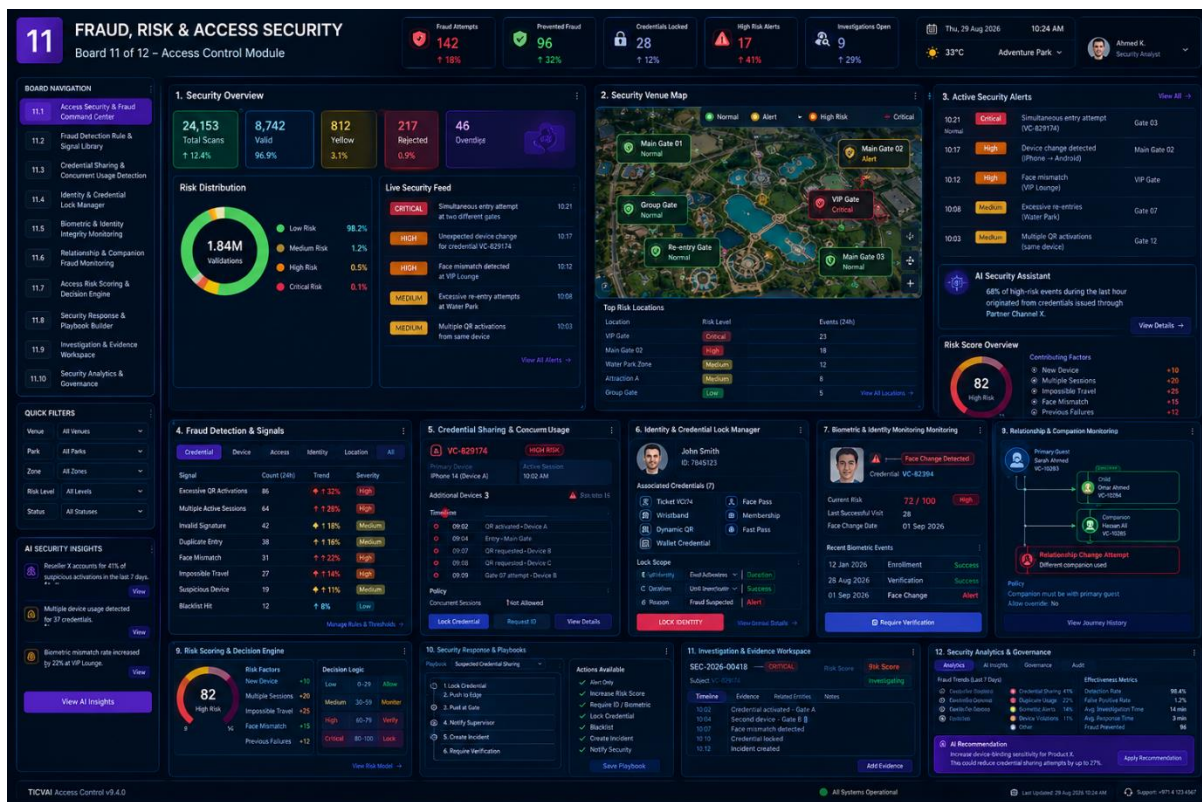
Gold Member + VIP Lounge + Private Event currently active + no Event Accreditation → DENY

That architecture prevents Board 10 from duplicating Board 2 and gives TICVAI a genuinely powerful enterprise **ABAC / contextual authorization engine** suitable for guests, memberships, employees, accreditations, restricted areas, events and multi-venue operations.

Access Control Module — Board 11

Fraud, Risk & Access Security

Board 11 of 12 — 10 Backend Configuration Screens



Board 11 is the real-time fraud prevention, credential abuse detection, identity protection, and security investigation layer of TICVAI Access Control.

By this stage:

- Board 2 decides access entitlement.
- Board 3 secures dynamic QR/digital credentials.

- **Board 5** manages biometric verification.
- **Board 7** protects offline access.
- **Board 9** manages live operational exceptions.
- **Board 10** evaluates contextual access policies.
- **Board 11** watches everything together to identify **fraudulent or suspicious behavior across credentials, identities, devices, gates, venues and sessions.**

The core principle is:

A credential may technically be valid while its usage behavior is suspicious. Board 11 determines the risk and security response.

Core Security Model

Credential Signals + Identity Signals + Device Signals + Scan Behavior + Location + Journey + Historical Activity

↓

TICVAI ACCESS RISK ENGINE

↓

Risk Score

↓

ALLOW / CHALLENGE / REVIEW / TEMPORARY LOCK / BLOCK / INVESTIGATE

Screen 11.1 — Access Security & Fraud Command Center

Purpose

Provide security teams with a real-time command center for access-related fraud and suspicious activity across all venues.

KPI Cards

Display:

- Active Security Alerts
- High-Risk Credentials
- Credentials Locked Today
- Suspicious QR Activity
- Device-Sharing Alerts
- Biometric Alerts
- Duplicate Access Attempts
- Blacklisted Credentials
- Active Investigations
- Fraud Prevented

Risk Distribution

Show:

- ☒ Low Risk
- ☐ Medium Risk

● High Risk

● Critical

Example:

1.84M validations today

Low: 98.2%

Medium: 1.2%

High: 0.5%

Critical: 0.1%

Live Security Feed

Examples:

CRITICAL

Credential VC-827194 attempted simultaneous entry at two different gates.

HIGH

Dynamic QR activated from four devices within 12 minutes.

HIGH

Face verification changed after previous successful admission.

MEDIUM

Unusually high re-entry attempts detected.

Security Venue Map

Display incidents geographically across:

Venue → Park → Zone → Gate

AI Security Summary

Example:

68% of high-risk events during the last hour originated from credentials issued through Partner Channel X.

Acceptance Condition

Security teams can understand the overall fraud and access-security posture from one screen.

Screen 11.2 — Fraud Detection Rule & Signal Library

Purpose

Configure the signals TICVAI uses to identify suspicious access behavior.

Credential Signals

Include:

- Excessive QR activations
- Multiple active sessions
- Credential copied
- Excessive refresh attempts
- Invalid signature
- Expired credential
- Revoked credential

- Screenshot/replay attempt
- Abnormal transfer frequency
- Repeated failed validation.

Device Signals

- New device
- Multiple devices
- Device-binding mismatch
- Rooted/compromised device where detectable
- abnormal device changes
- impossible device movement
- suspicious scanner/device activity.

Access Signals

- Duplicate entry
- simultaneous use
- anti-passback violations
- unusual re-entry
- unusual crossover
- excessive attraction use
- repeated wrong-gate attempts
- abnormal Fast Pass consumption.

Identity Signals

- Face mismatch
- unusual face change
- multiple identities linked
- suspicious companion changes
- POD/nanny relationship anomalies.

Location Signals

Example:

Credential scanned:

Gate A — 10:02

and

Gate B — 10:03

where physical travel between gates requires:

18 minutes

↓

IMPOSSIBLE TRAVEL SIGNAL

Signal Configuration

Each signal should support:

- Severity

- Weight
- Threshold
- Scope
- Applicable credential types
- Applicable venues
- Time window
- Offline availability
- Response.

Acceptance Condition

Fraud detection logic is configurable rather than hard-coded.

Screen 11.3 — Credential Sharing & Concurrent Usage Detection

Purpose

Detect one of the most important access-control fraud scenarios:

One valid ticket being shared by multiple people or devices.

Credential Session View

Example:

VC-829174

Primary Device:

iPhone — Device A

Current Session:

Active

Additional Devices:

3

Timeline

09:02 — QR activated Device A

09:04 — Main Gate entry

09:07 — QR requested Device B

09:08 — QR requested Device C

09:09 — Gate 07 attempt Device B

Detection Policies

Configure:

Maximum Active Devices

1

Maximum Device Changes

2 / Visit

Concurrent QR Sessions

Not Allowed

Simultaneous Gate Usage

Block

Response**Possible actions:**

- Increase Risk
- Require ID
- Require Biometric
- Require Operator
- Lock secondary device
- Lock credential
- Security alert.

Device Trust**Display:****Trusted Device****New Device****Unrecognized****Blocked****Acceptance Condition**

TICVAI detects credential sharing even when each individual QR or credential would otherwise pass technical validation.

Screen 11.4 — Unified Identity & Credential Lock Manager**Purpose**

Provide the **single security lock** required by the matrix so suspicious activity can immediately stop all access associated with an identity.

Identity

Guest: John Smith

Associated Credentials

- Ticket VC-18274
- RFID Wristband RF-8291
- Dynamic QR
- Wallet Credential
- Face Pass
- Membership
- Fast Pass.

Security Action

Instead of disabling each individually:

LOCK IDENTITY**Lock Scope****Select:**

- Credential only

- Media only
- Entitlement
- Venue
- All Venue Access
- Full Identity.

Lock Duration

- Until manually released
- End of day
- N hours
- Until investigation complete
- Permanent.

Lock Propagation

Show:

Central Platform ✓

Venue Edge ✓

Online Gates ✓

Offline Revocation Package ✓

Mobile Devices ✓

Reason

Examples:

- Credential Sharing
- Fraud Suspected
- Security Incident
- Identity Mismatch
- Stolen Credential
- Guest Removal.

Unlock

Should require appropriate permission and potentially dual authorization for high-risk locks.

Acceptance Condition

One security action can immediately restrict all configured access representations associated with a suspicious identity.

Screen 11.5 — Biometric & Identity Integrity Monitoring

Purpose

Detect suspicious biometric and identity-related changes without duplicating Board 5's biometric configuration.

Board 5 configures biometrics.

Board 11 monitors biometric security risk.

Security Events

Monitor:

- Face changed
- Re-enrollment
- repeated face mismatch
- multiple faces associated with one credential
- one face associated with multiple credentials
- suspicious enrollment frequency
- unusual verification failures.

Example**Credential:**

Annual Pass VC-82394

Original Face Enrollment:

12 Jan

Successful Visits:

28

Face Changed:

01 Sep

RISK ALERT**Reason:**

Significant identity change after established usage history

Historical Comparison**Show:**

Enrollment 1

↓

Enrollment 2

↓

Current verification.

Face Change Audit**Record:**

- Old biometric reference
- New biometric reference
- change date
- location
- operator
- verification process
- reason
- approval.

Actual biometric data should be handled according to applicable privacy/security requirements; the operational audit need not expose raw biometric templates.

Response**Possible:****Allow****Require ID****Supervisor Review****Security Investigation****Lock Credential****Acceptance Condition**

Identity changes and unusual biometric behavior automatically feed TICVAI's access-risk process.

Screen 11.6 — Relationship & Companion Fraud Monitoring

Purpose**Detect abuse involving linked guests such as:**

- Child + Adult
- POD + Companion
- Guest + Nanny
- Group Leader + Group
- Membership dependents.

Example**POD Credential:**

VC-POD-8271

Assigned Companion:

VC-CMP-2841

Today:

Companion A entered with primary guest.

Later:

Different Companion B attempted entry.

ALERT**COMPANION RELATIONSHIP CHANGE****Detection Scenarios****Configure:**

- Companion changed during visit
- Nanny credential used without primary guest
- Child enters/exits with unauthorized adult
- one companion linked to multiple active primary guests
- excessive relationship changes
- group leader used across unrelated groups.

Relationship Timeline

Display:**Primary Guest**

↓

Assigned Companion

↓

Entry

↓

Exit

↓

Relationship Change Attempt.

Child Protection**High-priority security alert for:****Child Exit + Assigned Adult Mismatch****Response**

- Yellow intervention
- Supervisor verification
- ID verification
- Biometric verification
- Security escalation
- Access denial.

Acceptance Condition

Linked-person relationships are continuously monitored rather than checked only at initial credential creation.

Screen 11.7 — Access Risk Scoring & Decision Engine

Purpose

Convert multiple security signals into a unified access risk score.

Risk Score**Example:****82 / 100****HIGH RISK****Contributing Signals**

Signal	Contribution
New Device	+10
Multiple Sessions	+20
Impossible Travel	+25

Signal	Contribution
Previous Failed Attempts	+12
Face Mismatch	+15

Risk Bands

Example:

0–29

☒ LOW

30–59

☐ MEDIUM

60–79

☒ HIGH

80–100

☒ CRITICAL

All thresholds should be configurable.

Response Matrix

Risk	Action
Low	Normal Validation
Medium	Enhanced Monitoring
High	Additional Verification
Critical	Lock + Security Review

Contextual Risk

Risk can depend on:

- Venue
- Product
- Ticket Value
- Event
- Access Zone
- Time
- Credential type
- historical behavior.

Example:

Risk Score 65

Normal Park Entry:

Require Verification

Restricted Backstage:**DENY****AI Risk Model**

TICVAI AI may identify combinations of behavior that static rules would not detect.

However:**AI ≠ Automatic Uncontrolled Blocking****Recommended architecture:****Rules + Risk Model + Policy → Controlled Security Decision****Acceptance Condition**

Multiple fraud indicators produce a consistent, explainable security risk assessment.

Screen 11.8 — Real-Time Security Response & Playbook Builder**Purpose**

Configure what TICVAI automatically does when security conditions are detected.

Example Playbook**SUSPECTED CREDENTIAL SHARING****Trigger:**

Concurrent Device Usage

AND

Risk Score > 75

Step 1

Temporarily Lock Credential

Step 2

Push Lock to Edge

Step 3

Show Red at Gate

Step 4

Notify Access Supervisor

Step 5

Create Security Incident

Step 6

Require ID/Biometric Verification

Available Actions

- Alert Only
- Increase Risk Score
- Require Additional Verification
- Require Supervisor
- Temporarily Lock

- Full Identity Lock
- Blacklist
- Notify Security
- Create Incident
- Notify Guest Services
- Trigger Edge Distribution.

Severity**Configure:****Informational****Low****Medium****High****Critical****Escalation****Example:****If incident not acknowledged within:**

3 minutes

→

Notify:**Security Supervisor****after:**

5 minutes

→

Notify:**Venue Operations Manager****Acceptance Condition**

Security responses are standardized, automated where appropriate and consistently executed across venues.

Screen 11.9 — Security Investigation & Evidence Workspace

Purpose

Provide security specialists with a deeper investigation environment than Board 9's operational incident workspace.

Board 9 handles:

“How do we resolve this guest problem now?”

Board 11 handles:

“Is this part of fraud, abuse or a larger security pattern?”

Investigation

SEC-2026-00418

Subject:

VC-829174

Risk:**CRITICAL — 91****Unified Timeline****Show:**

Ticket Purchased

↓

Credential Activated

↓

Device Bound

↓

Entry

↓

Second Device Activation

↓

Duplicate Scan

↓

Face Mismatch

↓

Credential Lock

↓

Operator Intervention.

Evidence**Automatically correlate:**

- Credential history
- Scan records
- Gate
- device IDs
- QR activations
- media changes
- transfer history
- biometric events
- companion relationships
- POS/ticket transaction reference
- overrides
- blacklist events
- security policies.

Related Entities

AI may identify:**5 credentials****sharing:**

Same Device

or:

Same suspicious behavior pattern

Investigation Graph**Visual relationship:****Device A**

⏪ ⏩ ⏴ ⏵

Credential 1

Credential 2

Credential 3

Investigator Actions

- Add Notes
- Add Evidence
- Link Incident
- Lock Identity
- Blacklist
- Clear Risk
- Escalate
- Close Investigation.

Case Status**Open**

→ Investigating

→ Action Taken

→ Resolved

→ Closed

Acceptance Condition

Security teams can reconstruct and investigate suspicious access behavior using a unified evidence trail.

Screen 11.10 — Security Analytics, AI Detection & Governance**Purpose**

Provide long-term intelligence on fraud patterns, security controls and effectiveness.

Fraud Analytics**Show:**

- Fraud Attempts
- Prevented Fraud
- Credential Sharing

- Duplicate Usage
- Biometric Alerts
- Device-Binding Violations
- Companion Violations
- Blacklist Hits
- Identity Locks
- Security Overrides.

Trend Analysis

Analyze by:

- Venue
- Park
- Event
- Product
- Channel
- Reseller
- Credential Type
- Media
- Device
- Gate
- Time/Day.

Example

Fraud Attempts by Sales Channel

Web: 8%

POS: 3%

B2B: 12%

Reseller X: 41%

Reseller Y: 7%

AI Pattern Detection

Example:

Emerging Pattern Detected

37 credentials purchased through the same reseller were activated across 64 different mobile devices and generated 112 duplicate-use attempts during the last seven days.

AI Recommendations

Examples:

Increase device-binding sensitivity for Product X.

Require additional verification for Reseller Y credentials above the configured risk threshold.

Review Gate 12 because its abnormal rejection pattern differs significantly from comparable gates.

Security Effectiveness

Measure:

- Detection Rate
- False Positive Indicator
- Operator Override Rate
- Average Investigation Time
- Average Response Time
- Recurring Fraud Rate
- Financial Exposure
- Estimated Fraud Prevented.

Governance

Security configuration lifecycle:

Draft → Test → Security Approval → Publish → Monitor → Review

Audit

Record changes to:

- fraud rules
- thresholds
- risk models
- response playbooks
- blacklist policies
- lock policies
- AI recommendations.

Acceptance Condition

TICVAI provides measurable security effectiveness and continuously identifies emerging fraud patterns.

Board 11 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
11.1	Access Security & Fraud Command Center	Real-time security posture
11.2	Fraud Detection Rule & Signal Library	Configure fraud indicators
11.3	Credential Sharing & Concurrent Usage Detection	Detect credential/device sharing
11.4	Unified Identity & Credential Lock Manager	Immediately restrict compromised identities
11.5	Biometric & Identity Integrity Monitoring	Detect biometric/identity anomalies
11.6	Relationship & Companion Fraud Monitoring	Protect child/POD/nanny/group relationships
11.7	Access Risk Scoring & Decision Engine	Calculate explainable security risk

#	Backend Screen	Main Responsibility
11.8	Real-Time Security Response & Playbook Builder	Automate controlled security responses
11.9	Security Investigation & Evidence Workspace	Investigate fraud and correlate evidence
11.10	Security Analytics, AI Detection & Governance	Analyze patterns and improve controls

Board 11 Key Workflow

Collect Signals → Detect Suspicious Behavior → Correlate Identity/Credential/Device → Calculate Risk → Determine Security Response → Challenge or Lock → Alert Security → Investigate → Resolve → Analyze Pattern → Improve Controls

Critical separation from the other boards

This distinction will keep the module clean:

Board 3 — Credential Security

Protects how the credential itself works.

Board 5 — Biometric Access

Configures how biometric identity is enrolled and verified.

Board 9 — Live Operations

Helps the operator resolve today's access exception.

Board 10 — Dynamic Policy

Determines whether access is permitted under current contextual conditions.

Board 11 — Fraud & Risk

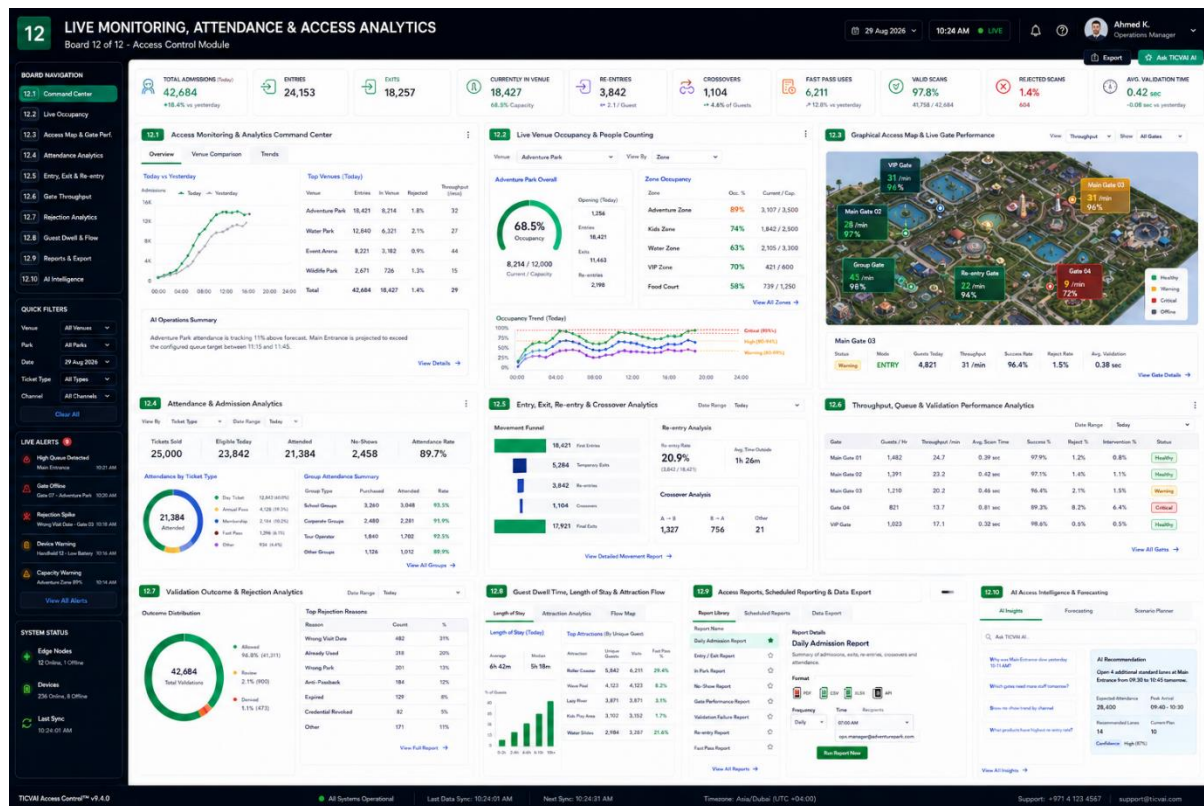
Determines whether otherwise valid activity is suspicious, abusive, compromised or fraudulent.

That makes Board 11 effectively the **security brain sitting above the entire TICVAI Access Control ecosystem**, rather than another blacklist or invalid-ticket screen.

Access Control Module — Board 12

Live Monitoring, Attendance & Access Analytics

Board 12 of 12 — 10 Backend Configuration Screens



Board 12 is the **monitoring, attendance intelligence, operational analytics, and management reporting** layer of the TICVAI Access Control module.

It closes the complete Access Control architecture by answering:

What is happening across the venue right now, what happened historically, where are guests located, how efficiently are gates operating, and what should operations improve?

The board consolidates access events generated by QR, barcode, RFID, NFC, Face Pass/Face Tag, turnstiles, handhelds, external access-control systems, entry/exit/re-entry/crossover events, Fast Pass validations, group admissions, and offline transactions into a unified operational analytics model.

A key distinction:

Board 9 = operators take live operational action.

Board 12 = management monitors, analyzes, reports, forecasts and optimizes the operation.

Screen 12.1 — Access Monitoring & Analytics Command Center

Purpose

Provide a single executive and operational overview of access performance across all TICVAI-controlled venues.

Global KPI Cards

Show:

- Total Admissions Today

- Entries
- Exits
- Currently In Venue
- Re-entries
- Crossovers
- Group Admissions
- Fast Pass Uses
- Valid Scans
- Rejected Scans
- Intervention Rate
- Average Validation Time
- Active Gates
- Offline Devices

Example**TOTAL ADMISSIONS**

42,684

CURRENTLY IN VENUE

18,427

VALIDATION SUCCESS

97.8%

AVG. VALIDATION

0.42 sec

Venue Comparison

Venue	Entries	In Venue	Rejected	Throughput
Adventure Park	18,421	8,214	1.8%	32/min
Water Park	12,840	6,321	2.1%	27/min
Event Arena	8,221	3,182	0.9%	44/min

AI Operations Summary**Example:**

Adventure Park attendance is tracking 11% above forecast. Main Entrance is projected to exceed the configured queue target between 11:15 and 11:45.

Acceptance Condition

Management can understand the overall access operation from one screen.

Screen 12.2 — Live Venue Occupancy & People Counting

Purpose

Provide real-time people counting and occupancy using entry and exit events.

Occupancy Equation

Opening Occupancy

•

Entries

–

Exits

±

Operational Adjustments

=

CURRENT OCCUPANCY

Hierarchical View

Show occupancy by:

Venue

→ Park

→ Zone

→ Attraction

→ Controlled Area.

Example

Adventure Park

Current:

8,214

Capacity:

12,000

Occupancy:

68.5%

Zone Status

Kids Zone:

1,842 / 2,500

 74%

Adventure Zone:

3,107 / 3,500

 89%

VIP Zone:

421 / 600

 70%

Thresholds

Administrators configure:**Normal**

0–79%

Warning

80–89%

High

90–94%

Critical

95%+

Alerts**Example:**

Adventure Zone has exceeded its 90% operational threshold.

Acceptance Condition

TICVAI provides an accurate current in-venue/in-zone population based on access activity.

Screen 12.3 — Graphical Access Map & Live Gate Performance**Purpose**

Turn the graphical access topology created in Board 1 into a live operational analytics map.

Venue Digital Twin**Display:**

- Gates
- Turnstiles
- Entry points
- Exit points
- Re-entry gates
- Group gates
- VIP gates
- attraction access
- crossover points.

Status

-  Healthy
-  Warning
-  Critical
-  Offline

Gate Overlay**Example:****MAIN GATE 03****Guests:**

4,821

Throughput:

31/min

Success:

96.4%

Reject:

2.1%

Yellow:

1.5%

Average Validation:

0.38 sec

Heatmap

Visualize:

Guest Flow

Queue Pressure

Rejection Rate

Device Health

Occupancy

Drill Down

Venue → Park → Zone → Access Point → Gate → Device

Acceptance Condition

Operations can visually identify where access bottlenecks or abnormal conditions are occurring.

Screen 12.4 — Attendance & Admission Analytics

Purpose

Provide detailed reporting of who actually attended compared with tickets sold/reserved.

This is particularly important because group admission may differ from purchased quantity.

Attendance KPIs

Show:

- Tickets Sold
- Tickets Eligible Today
- Tickets Scanned
- Unique Guests
- No-Shows
- Group Attendance
- Membership Attendance
- Repeat Entry
- Attendance Rate.

Example**Sold:**

25,000

Eligible:

23,842

Attended:

21,384

ATTENDANCE RATE

89.7%

Breakdown**Analyze by:**

- Ticket Type
- Product
- Event
- Timeslot
- Membership
- Channel
- B2B Partner
- Reseller
- Customer Segment
- Guest Category
- Venue.

Group Analytics**Example:****School Group:****Purchased:**

120

Actual Attendance:

112

Attendance:

93.3%

No-Show Analysis**Show:**

- no-show rate
- ticket type
- channel
- event
- date/time
- customer segment.

AI Insight

Morning school groups average 94% attendance, while afternoon groups average 81%. Consider adjusting capacity assumptions for afternoon group products.

Acceptance Condition

TICVAI distinguishes purchased capacity from actual physical attendance.

Screen 12.5 — Entry, Exit, Re-entry & Crossover Analytics**Purpose**

Analyze complete guest movement across the access journey.

Journey Metrics**Show:**

First Entry

Exit

Re-entry

Crossover

Final Exit

Journey Funnel

Example:

18,421 First Entries

↓

5,284 Temporary Exits

↓

3,842 Re-entries

↓

1,104 Crossovers

↓

17,921 Final Exits

Re-entry Analytics**Measure:**

- Re-entry rate
- average time outside
- most-used re-entry gates
- rejected re-entry
- re-entry by product.

Crossover Analytics**Measure:**

- Park A → Park B
- Park B → Park A
- crossover time

- crossover product
- crossover utilization.

Multi-Park Flow

Example:

Adventure Park

2,184

→

Water Park

1,327

Acceptance Condition

Management can understand actual movement patterns rather than only total admission counts.

Screen 12.6 — Throughput, Queue & Validation Performance Analytics

Purpose

Measure the operational efficiency of gates and validation devices.

Gate KPIs

Show:

- Guests per Minute
- Guests per Hour
- Average Scan Time
- Average Gate Cycle
- Success Rate
- Yellow Rate
- Reject Rate
- Manual Intervention Rate
- Downtime.

Comparison

Gate	Guests/hr	Validation	Reject	Intervention
G01	1,482	0.39s	1.2%	0.8%
G02	1,391	0.42s	1.4%	1.1%
G03	821	0.81s	8.2%	6.4%

Bottleneck Detection

TICVAI highlights:

GATE 03 — UNDERPERFORMING

Potential reasons:

- QR read failures

- excessive manual verification
- hardware latency
- policy complexity
- wrong guest routing.

Queue Analytics

Show:

Current

Historical

Forecast

AI Recommendation

Gate 03 processes 39% fewer guests per minute than comparable gates. RFID read retries are the primary contributor.

Acceptance Condition

Management can identify why specific gates or lanes are underperforming.

Screen 12.7 — Validation Outcome & Rejection Analytics

Purpose

Analyze why guests are denied or require manual intervention.

Outcome Distribution

☒ Allowed

96.8%

☐ Operator Review

2.1%

☒ Denied

1.1%

Rejection Reasons

Example:

Reason	Count	%
Wrong Visit Date	482	31%
Already Used	318	20%
Wrong Park	201	13%
Anti-Passback	184	12%
Expired	129	8%
Credential Revoked	82	5%

Reason	Count	%
Other	171	11%

Analyze By

- Venue
- Gate
- Product
- Ticket Type
- Channel
- Reseller
- Operator
- Device
- Credential Type
- Time.

Override Correlation**Example:****Wrong Visit Date****Rejected:**

482

Overrides:

281

Override Rate:**58.3%**

This may indicate a configuration/business-process problem rather than fraud.

AI Root Cause

72% of Wrong Visit Date overrides originate from tickets sold through Reseller X. Review the reseller's date-mapping configuration.

Acceptance Condition

Management can identify the operational root cause behind access failures.

Screen 12.8 — Guest Dwell Time, Length of Stay & Attraction Flow**Purpose**

Use access events to understand how guests move through and use the venue.

Length of Stay**Calculate where sufficient entry/exit data exists:****Entry**

09:12

Final Exit

16:42

Length of Stay

7h 30m

Analytics**Show:**

- Average Length of Stay
- Median Stay
- Peak Arrival
- Peak Departure
- Zone Dwell Time
- Attraction Visits
- Fast Pass Usage
- Re-entry behavior.

Guest Flow**Visualize:****Main Entrance**

↓

Adventure Zone

↓

Coaster

↓

F&B

↓

Water Zone

↓

Exit

Attraction Analytics**Example:****Roller Coaster****Unique Guests:**

5,842

Total Validations:

6,211

Fast Pass:

1,827

Repeat Visits:

369

Peak:

14:00–15:00

Privacy Principle

Analytics should primarily use pseudonymized/aggregated operational identifiers where individual identity is unnecessary.

Acceptance Condition

TICVAI can analyze venue utilization from access events without requiring separate manual attendance systems.

Screen 12.9 — Access Reports, Scheduled Reporting & Data Export

Purpose

Provide configurable operational and management reports.

Report Library**Include:**

Daily Admission Report

Entry / Exit Report

In-Park Report

Attendance Report

No-Show Report

Gate Performance Report

Validation Failure Report

Re-entry Report

Crossover Report

Fast Pass Report

Group Attendance Report

Device Performance Report

Override Report

Offline Access Report

Access Security Summary

Report Builder

Filters:

- Tenant
- Venue
- Park
- Zone
- Event
- Date
- Ticket Type
- Product
- Gate
- Device
- Channel

- Partner.

Schedule

Example:

Daily Access Summary

Frequency:

Daily

Time:

07:00

Recipients / destination:

Configured authorized recipients or reporting destinations.

Formats

Support as appropriate:

- Dashboard
- CSV
- XLSX
- PDF
- API/Data Feed
- BI integration.

Data Governance

Apply:

- RBAC
- tenant isolation
- field-level restrictions
- retention rules
- export audit.

Acceptance Condition

Authorized users can create, schedule and export access-control reports without development support.

Screen 12.10 — AI Access Intelligence, Forecasting & Executive Insights

Purpose

Turn access-control data into proactive operational intelligence.

This should be the **final intelligence screen of the entire Access Control module**.

AI Ask TICVAI

Provide natural-language analytics.

Example:

Why was Main Entrance slow yesterday between 10 AM and 11 AM?

TICVAI:

Throughput decreased 22%. Gate 04 was offline for 18 minutes and Gate 07 experienced elevated QR retries. Two school groups totaling 286 guests also arrived within the same 12-minute period.

Forecasting

Forecast:

- Tomorrow's Attendance
- Peak Arrival Time
- Peak Exit Time
- Venue Occupancy
- Zone Occupancy
- Gate Demand
- Group Arrival Pressure
- Re-entry Demand
- Device Capacity.

Example

TOMORROW FORECAST

Expected Attendance:

28,400

Peak Arrival:

09:40–10:30

Recommended Entry Lanes:

14

Current Planned:

10

AI Recommendation

Open four additional standard lanes from 09:30–10:45 and temporarily allocate Group Gate 02 to B2B arrivals between 10:00–10:30.

Scenario Planning

Operations can ask:

What happens if attendance reaches 35,000?

AI estimates:

- gate requirements
- throughput
- occupancy risk
- queue duration
- device utilization.

Executive Insights

Show:

Attendance vs Forecast

Guest Flow Efficiency

Gate Efficiency

Access Failure Rate

Security Impact**Venue Utilization****Operational Recommendations****Human Control****AI should:****Analyze → Explain → Forecast → Recommend**

Operational actions remain governed by the appropriate approval/permission controls.

Acceptance Condition

TICVAI turns access data into explainable forecasts and actionable recommendations rather than providing dashboards alone.

Board 12 — Final 10-Screen Structure

#	Backend Screen	Main Responsibility
12.1	Access Monitoring & Analytics Command Center	Executive access overview
12.2	Live Venue Occupancy & People Counting	Real-time occupancy
12.3	Graphical Access Map & Live Gate Performance	Visual venue/gate monitoring
12.4	Attendance & Admission Analytics	Actual attendance and no-shows
12.5	Entry, Exit, Re-entry & Crossover Analytics	Complete guest movement
12.6	Throughput, Queue & Validation Performance Analytics	Gate operational efficiency
12.7	Validation Outcome & Rejection Analytics	Failure and intervention analysis
12.8	Guest Dwell Time, Length of Stay & Attraction Flow	Guest movement/utilization
12.9	Access Reports, Scheduled Reporting & Data Export	Reporting and BI distribution
12.10	AI Access Intelligence, Forecasting & Executive Insights	AI-driven operational intelligence

Board 12 Key Workflow

Collect Access Events → Normalize → Calculate Attendance → Calculate Occupancy → Analyze Guest Movement → Measure Gate Performance → Identify Failures → Analyze Venue Utilization → Report → Forecast → AI Recommend

Final Access Control Architecture

With Board 12, the **12-board / 120-screen Access Control module is complete:**

Board	Area
1	Access Control Command Center & Venue Topology

Board	Area
2	Access Rules & Entitlement Engine
3	Dynamic QR & Secure Digital Credentials
4	Media, Credential & Verification Methods
5	Biometric Access — Face Pass & Face Tag
6	Devices, Turnstiles & Gate Configuration
7	Offline Access & Edge Operations
8	Guest Journeys, Groups & Special Admission
9	Live Access Operations, Podium & Exception Management
10	Dynamic Policy & Attribute-Based Access Control
11	Fraud, Risk & Access Security
12	Live Monitoring, Attendance & Access Analytics

The separation is important because **Board 12 should not become another general TICVAI BI module**. It owns the **access-control operational analytics domain** and can feed TICVAI's wider reporting/Power BI/AI analytics layer with normalized access data. That keeps the Access Control module complete while avoiding duplication with the platform-wide analytics capability.